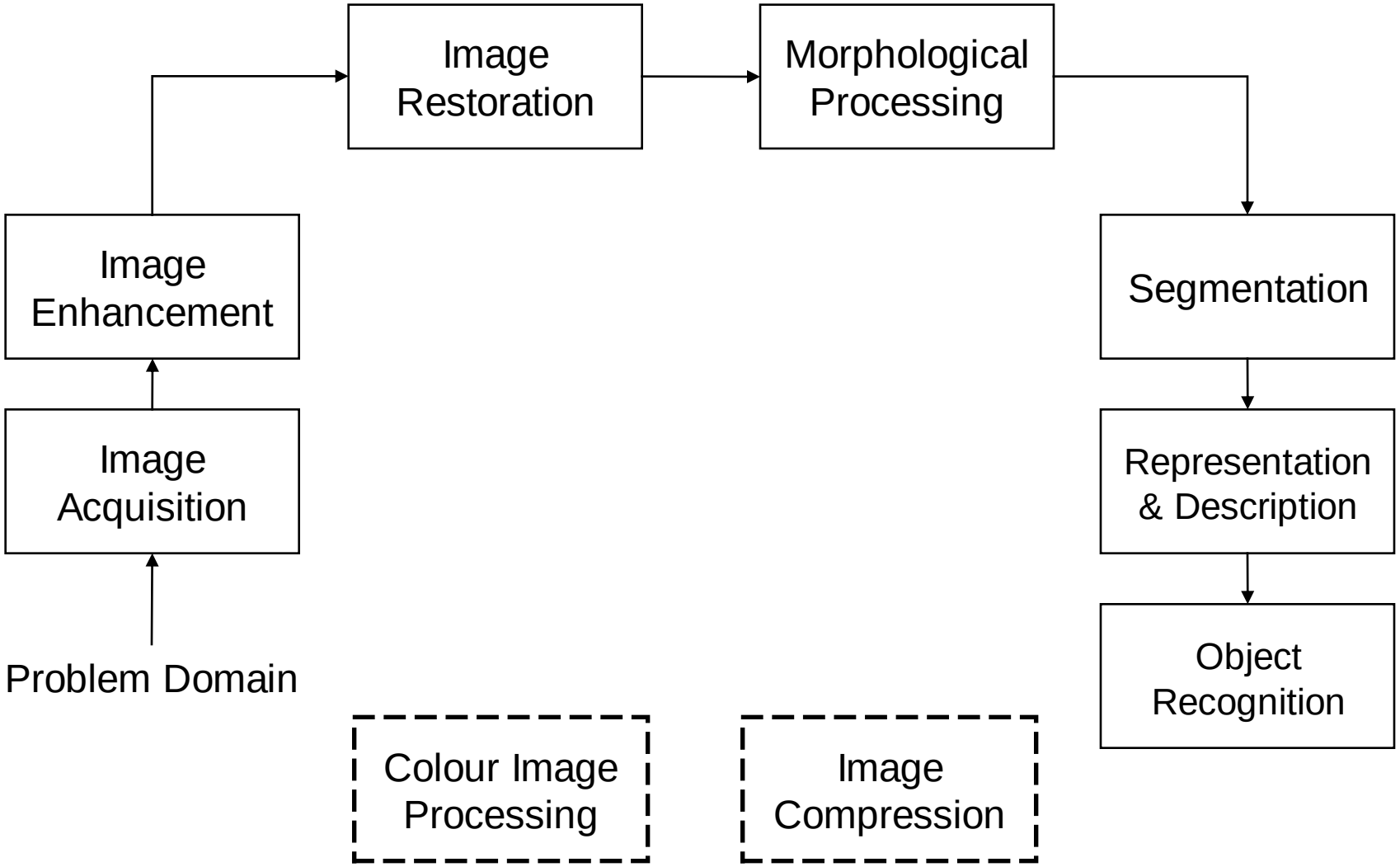
Image Morphology

Content derived from multiple sources

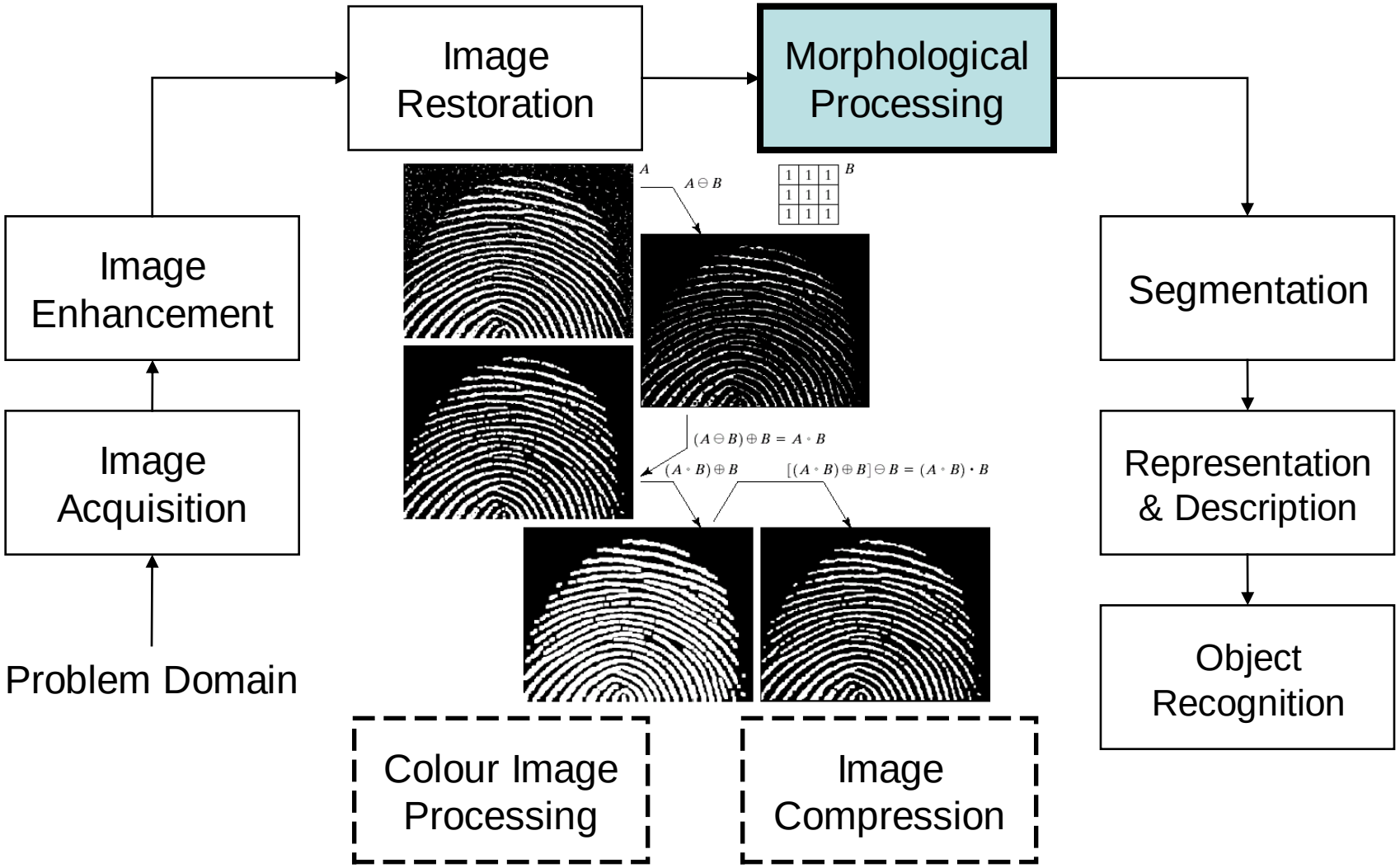
Contents

- ✓ What is Morphology?
- ✓ Fundamental Operations
- ✓ Morphological Algorithms
- ✓ Mathematical Examples

Phases of Digital Image Processing



Phases of Digital Image Processing : Morphological Processing



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

What is Morphology?

Morphological image processing (or *morphology*) describes a range of image processing techniques that deal with the shape (or morphology) of features in an image.

Morphological operations are typically applied to remove imperfections introduced during segmentation, and so typically operate on bi-level images.

Example



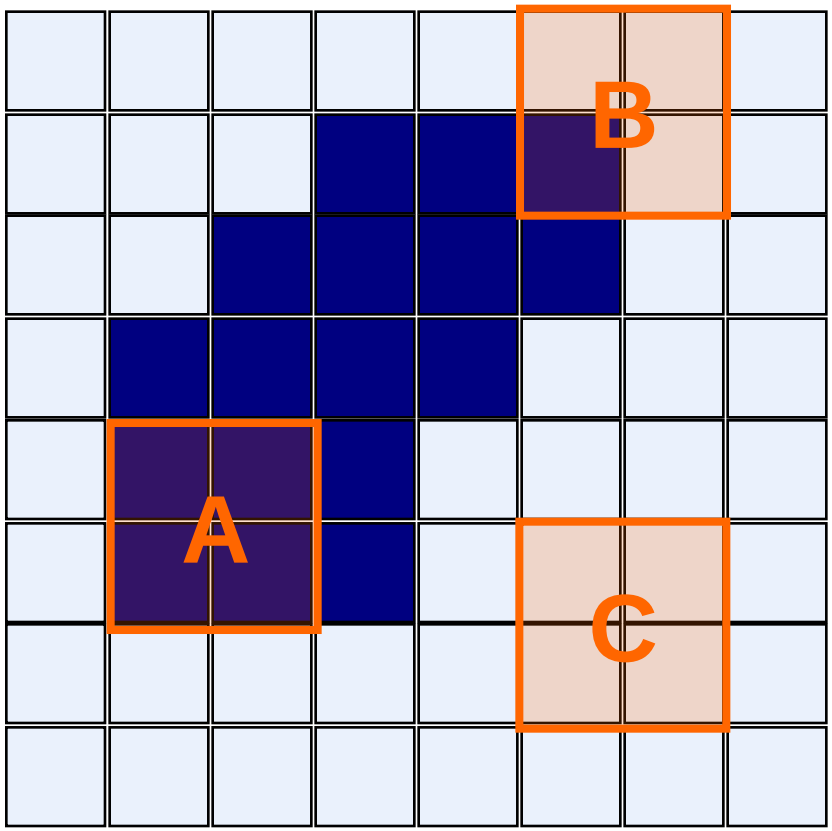
Image after segmentation

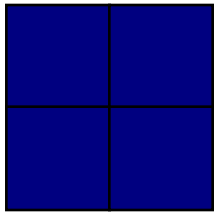


Image after segmentation and morphological processing

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Structuring Elements, Hits & Fits





Structuring Element

Fit: All *on pixels* in the structuring element cover *on pixels* in the image.

Hit: Any *on pixel* in the structuring element covers an *on pixel* in the image.

All morphological processing operations are based on these simple ideas.

Structuring Elements

Structuring elements can be any size and make any shape.

For simplicity we use rectangular structuring elements with their origin at the middle pixel.

1	1	1
1	**1**	1
1	1	1

0	1	0
1	**1**	1
0	1	0

0	0	1	0	0
0	1	1	1	0
1	1	**1**	1	1
0	1	1	1	0
0	0	1	0	0

Fitting & Hitting

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0	0	0
0	0	1	B	1	1	1	0	C	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	A	1	1	1	0
0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0

1	1	1
1	1	1
1	1	1

Structuring Element 1

0	1	0
1	1	1
0	1	0

Structuring Element 2

Fitting & Hitting

Image Region	Structuring Element 1		Structuring Element 2	
	Fit	**Hit**	**Fit**	**Hit**
A	YES	YES	YES	YES
B	NO	YES	YES	YES
C	NO	YES	NO	NO

Fundamental Operations

The structuring element is moved across every pixel in the original image to give a pixel in a new processed image.

The value of this new pixel depends on the operation performed.

There are two basic morphological operations: erosion and dilation.

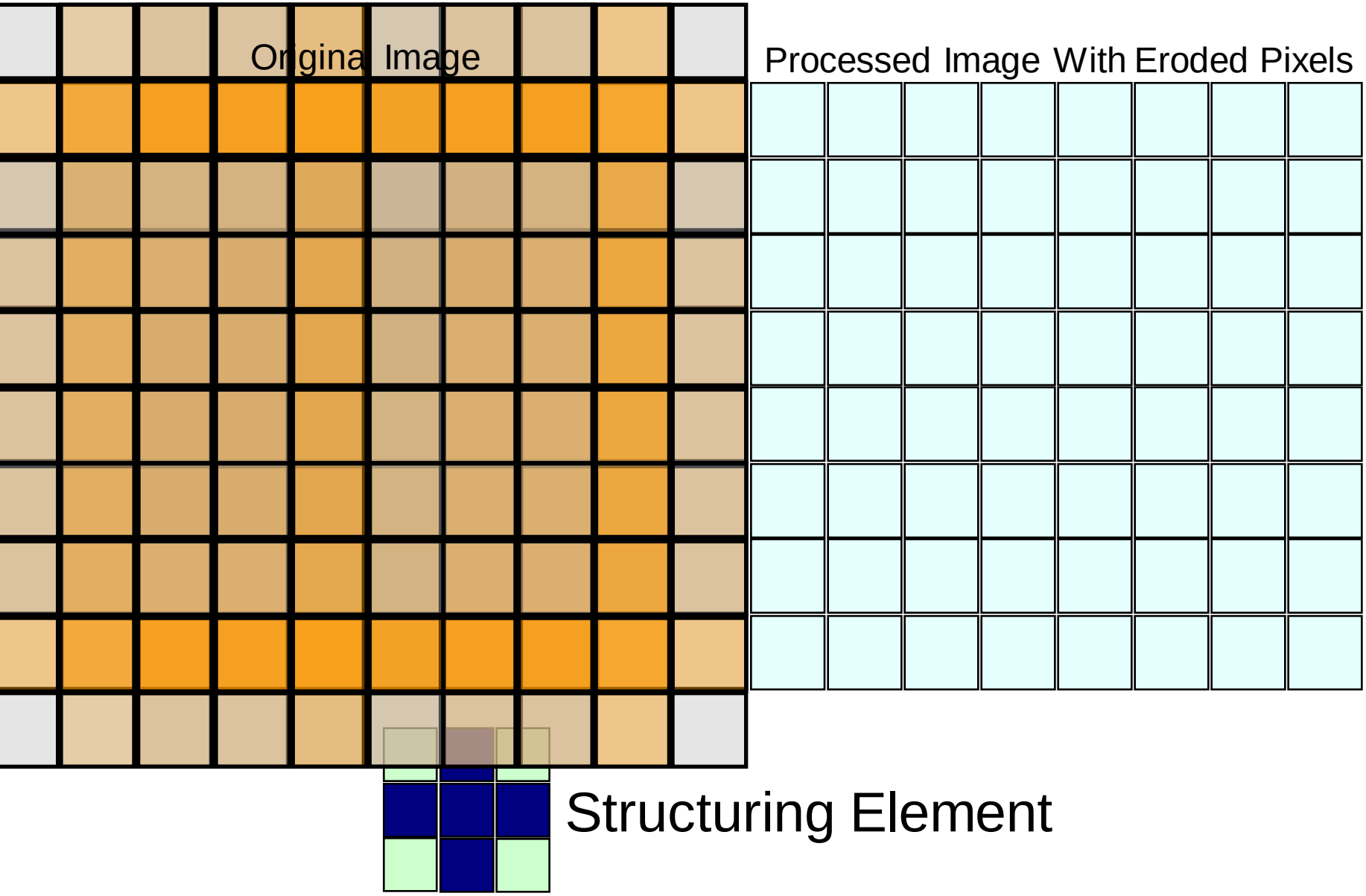
Erosion

Erosion of image *f* by structuring element *s* is given by $f \ominus s$.

The structuring element s is positioned with its origin at *(r, c)* and the new pixel value is determined using the rule:

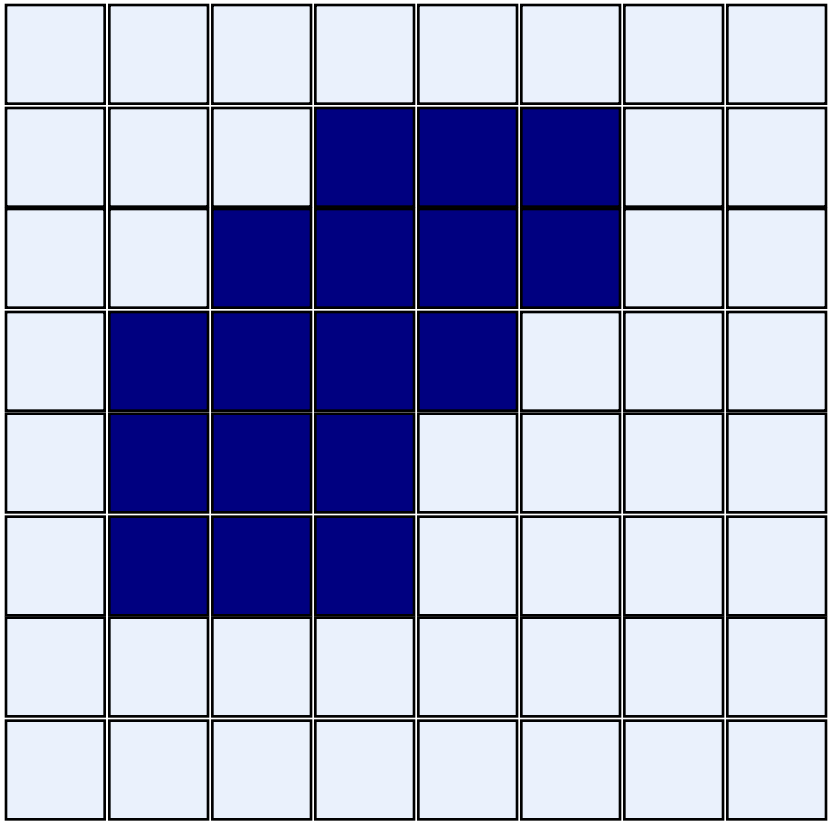
$$g(r,c) = \begin{cases} 1 & \text{if } s \text{ fits } f \\ 0 & \text{otherwise} \end{cases}$$

Erosion Example

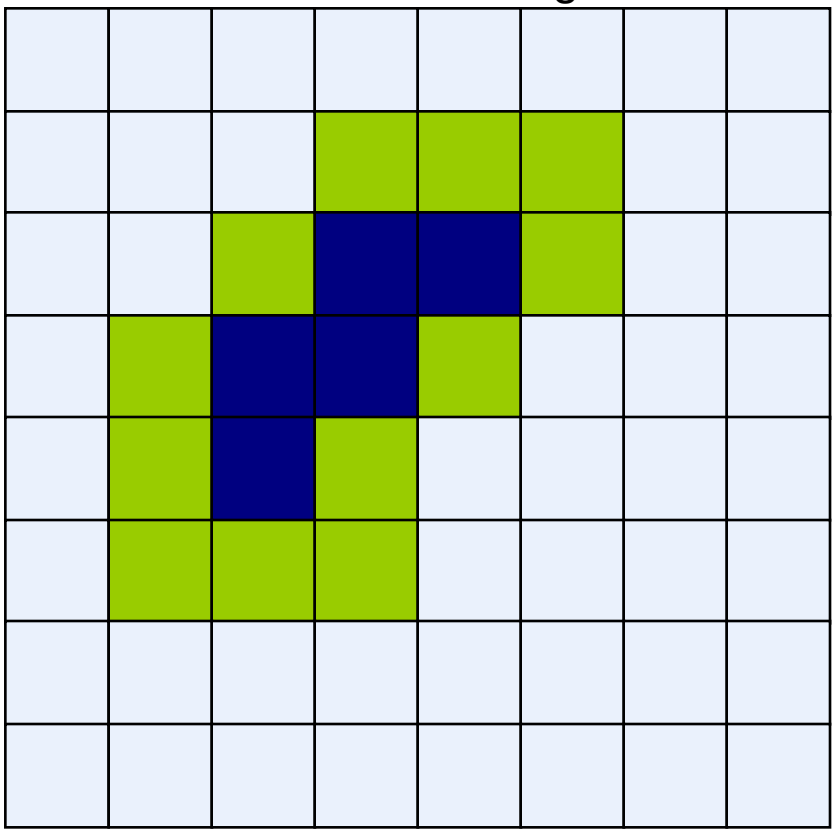


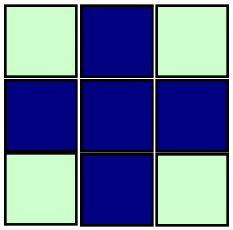
Erosion Example

Original Image



Processed Image



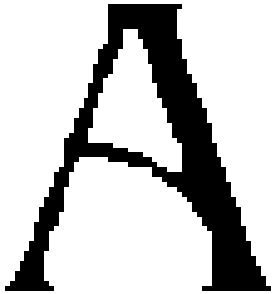


Structuring Element

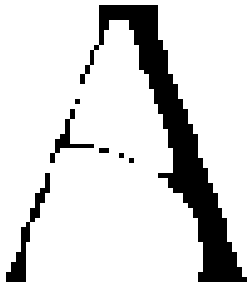
Erosion Example 1



Original image

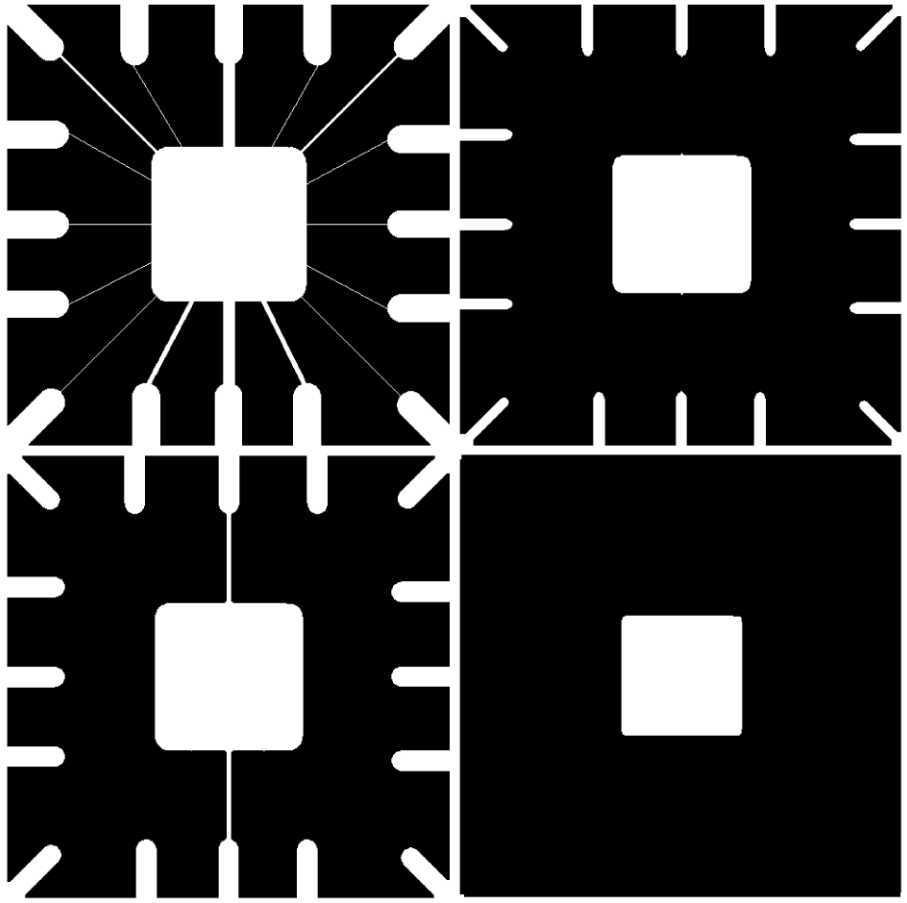


Erosion by 3*3 square structuring element



Erosion by 5*5 square structuring element

Erosion Example 2



Original image

After erosion with a disc of radius 10

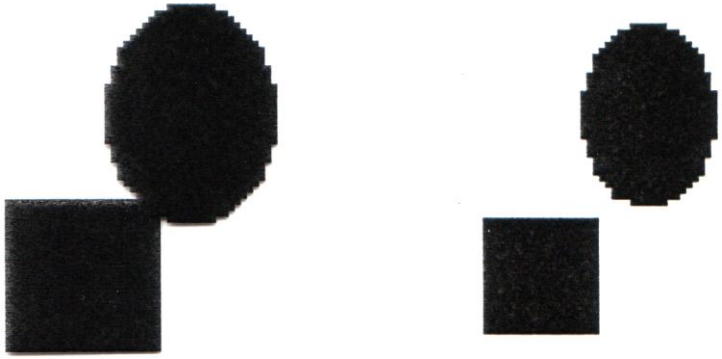
After erosion with a disc of radius 5

After erosion with a disc of radius 20

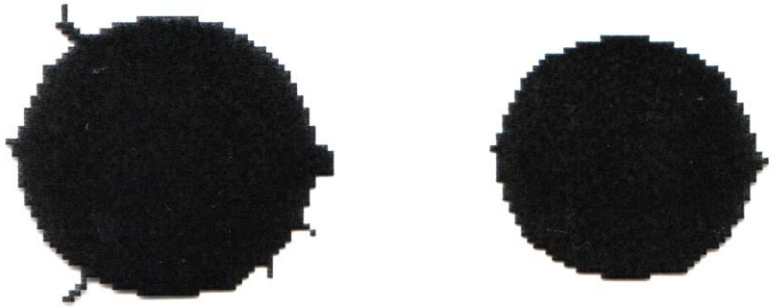
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Use of Erosion

Erosion can split apart joined objects



Erosion shrinks objects



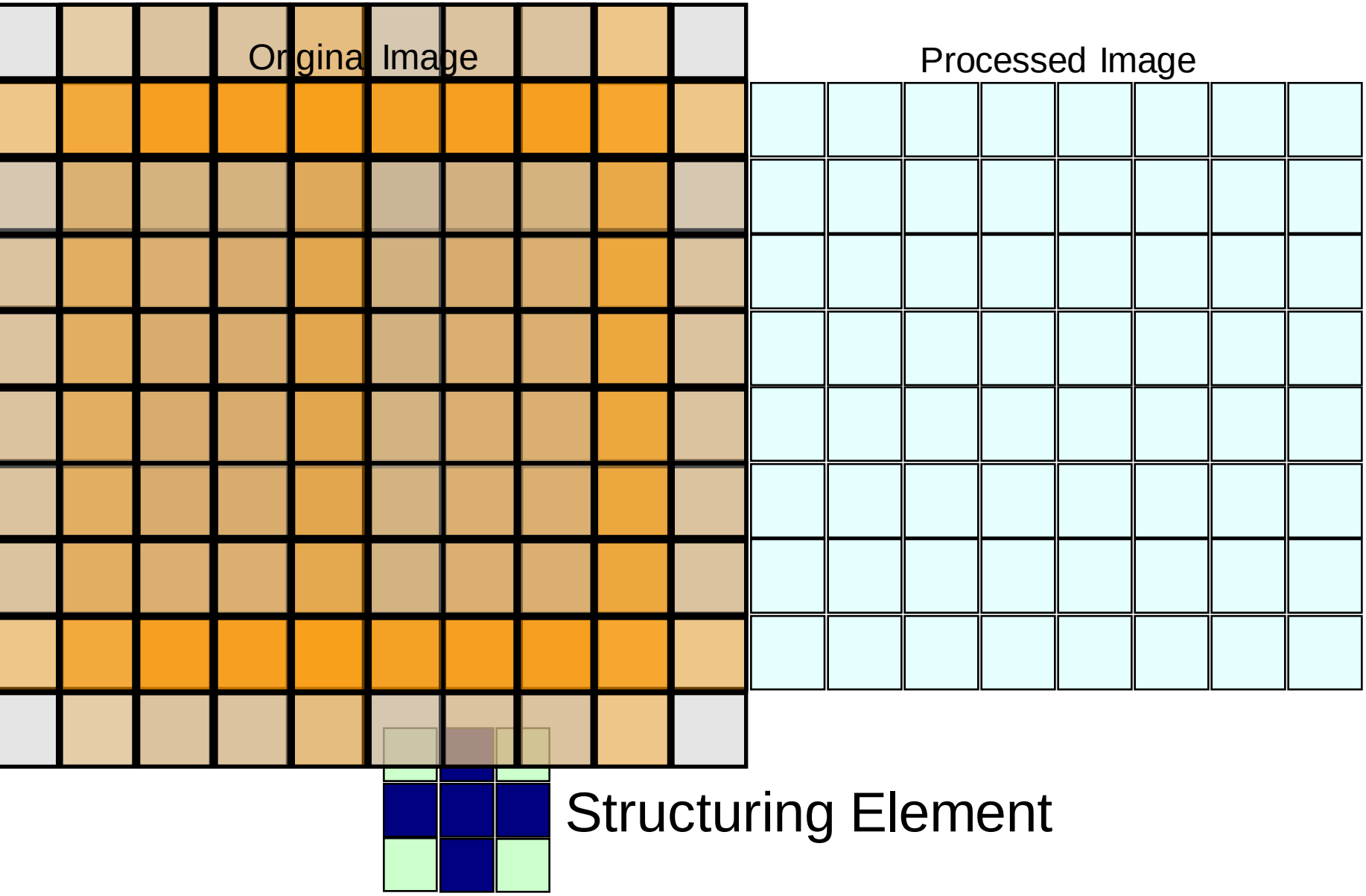
Dilation

Dilation of image *f* by structuring element *s* is given by $f \oplus s$.

The structuring element s is positioned with its origin at *(r, c)* and the new pixel value is determined using the rule:

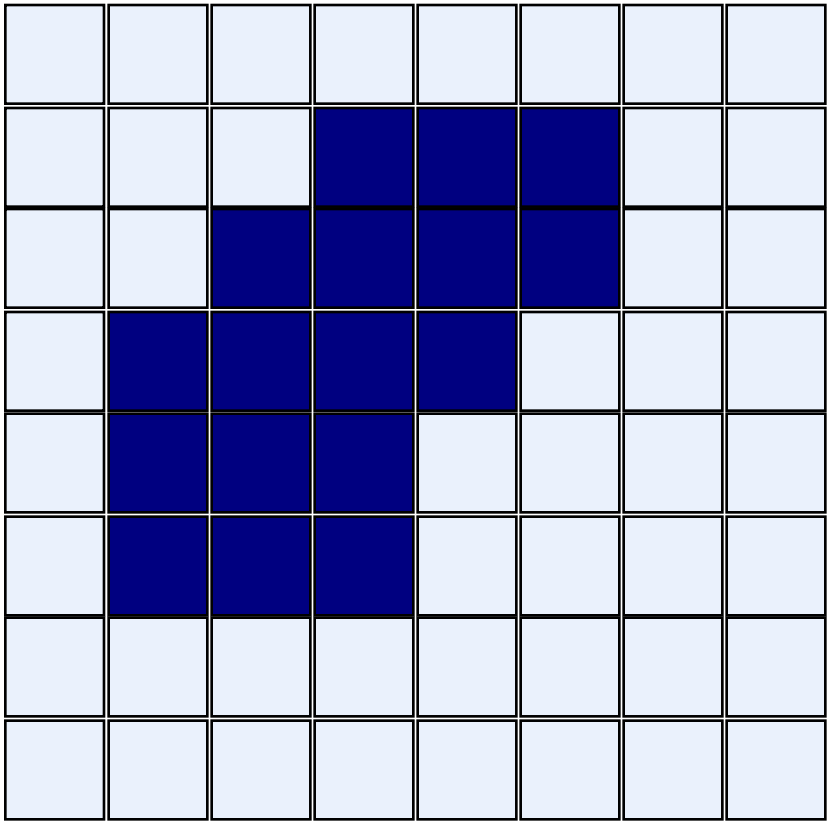
$$g(r,c) = \begin{cases} 1 & \text{if } s \text{ hits } f \\ 0 & \text{otherwise} \end{cases}$$

Dilation Example

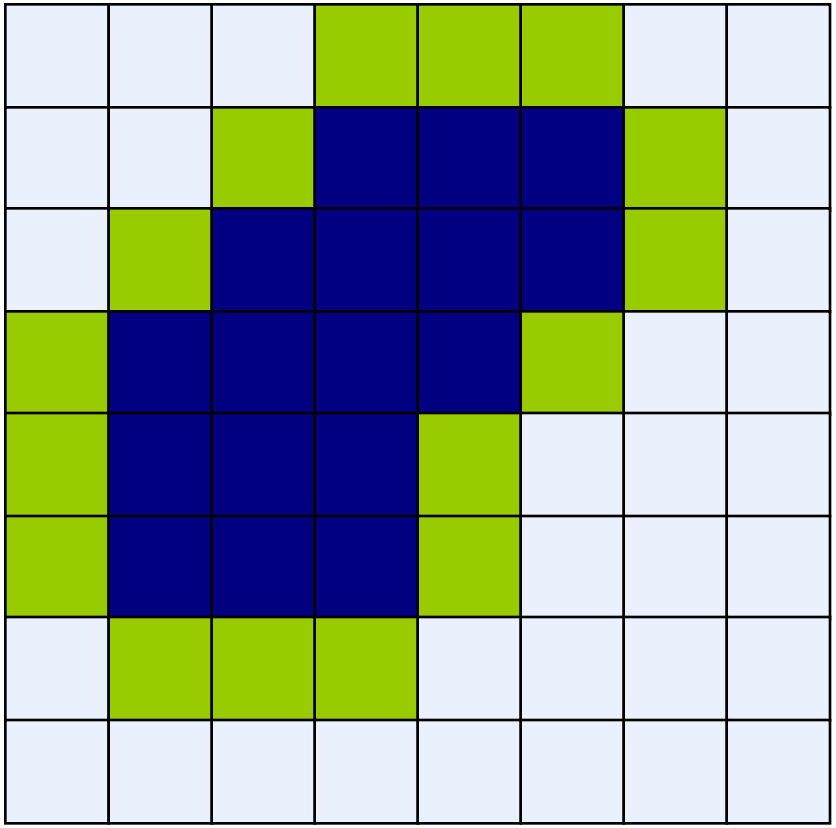


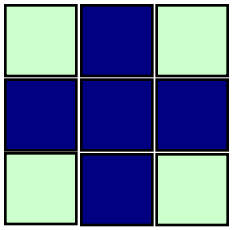
Dilation Example

Original Image



Processed Image With Dilated Pixels





Structuring Element

Dilation Example 1



Original image



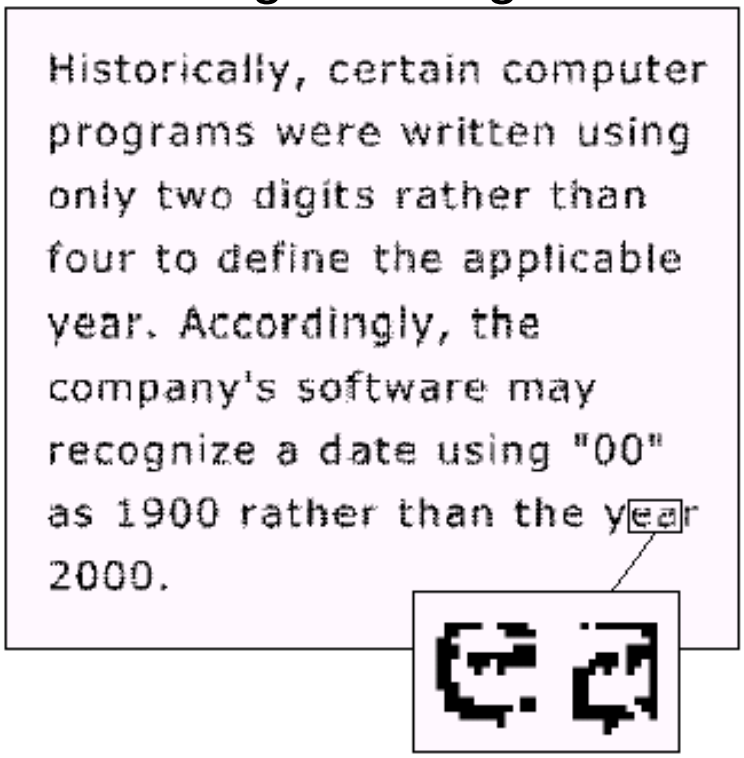
Dilation by 3*3 square structuring element



Dilation by 5*5 square structuring element

Dilation Example 2

Original image



After dilation



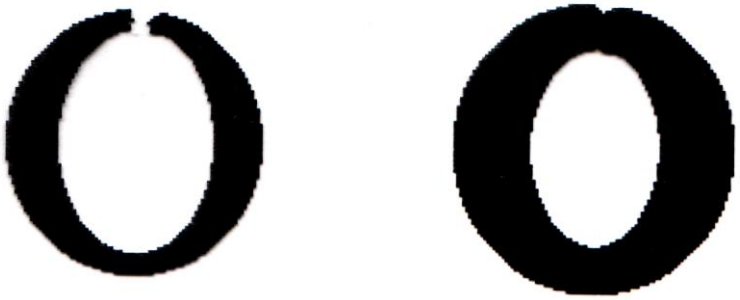
0	1	0
1	1	1
0	1	0

Structuring element

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Use of Dilation

Dilation can repair breaks



Dilation enlarges objects



Compound Operations

More effective morphological operations can be performed by performing combinations of erosions and dilations.

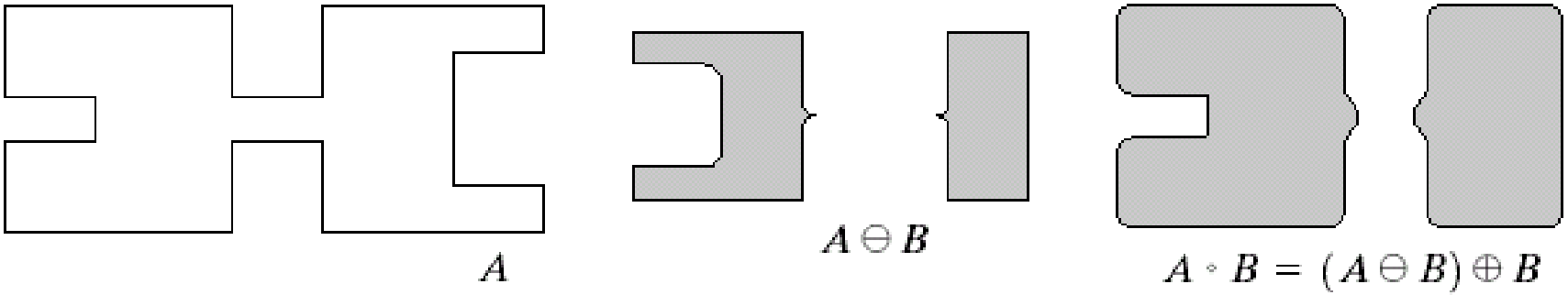
The most widely used of these *compound operations* are:

- Opening
- Closing

Opening

The opening of image *f* by structuring element *s*, denoted $f \circ s$ is simply an <u>erosion followed by a dilation.</u>

$$f \circ s = (f \ominus s) \oplus s$$



Original shape | After erosion | After dilation (opening)

A disc shaped structuring element is used.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Opening Example

Original Image

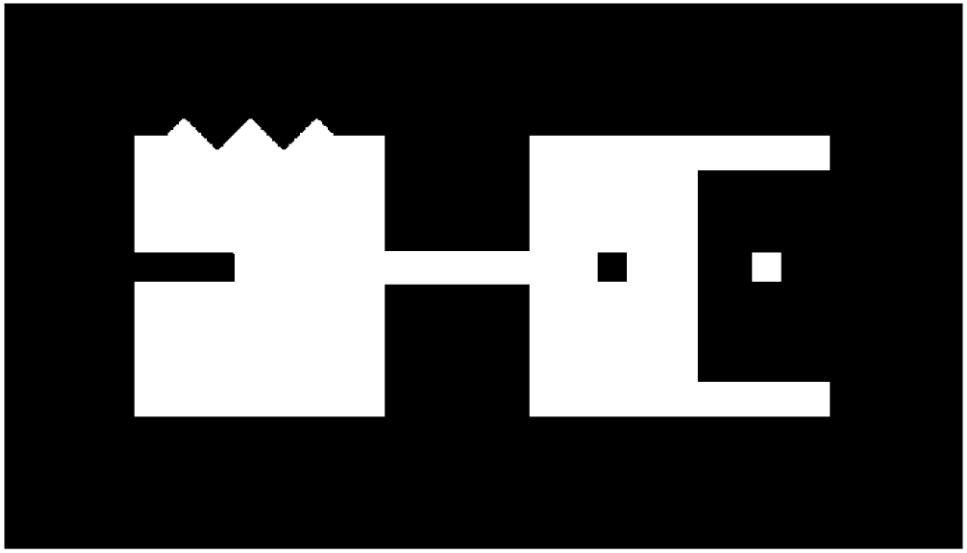
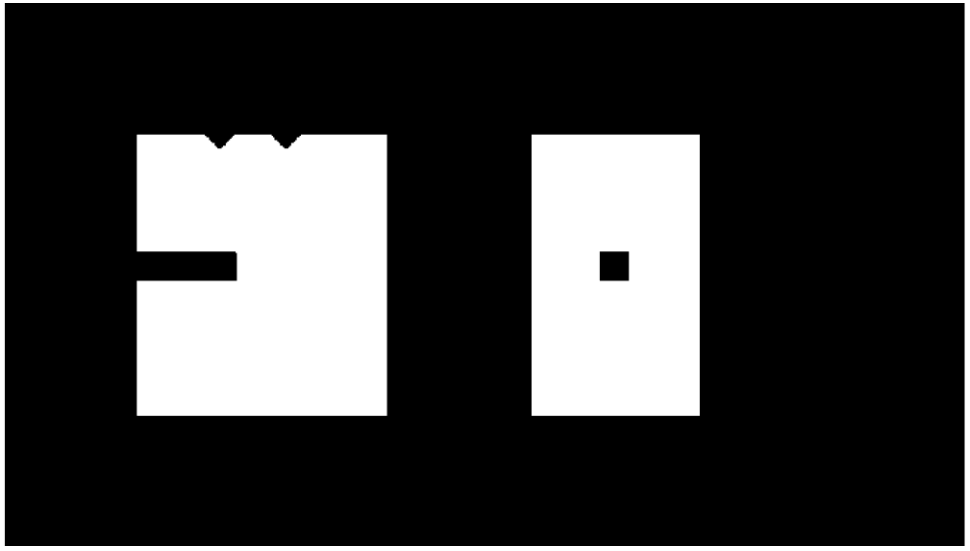
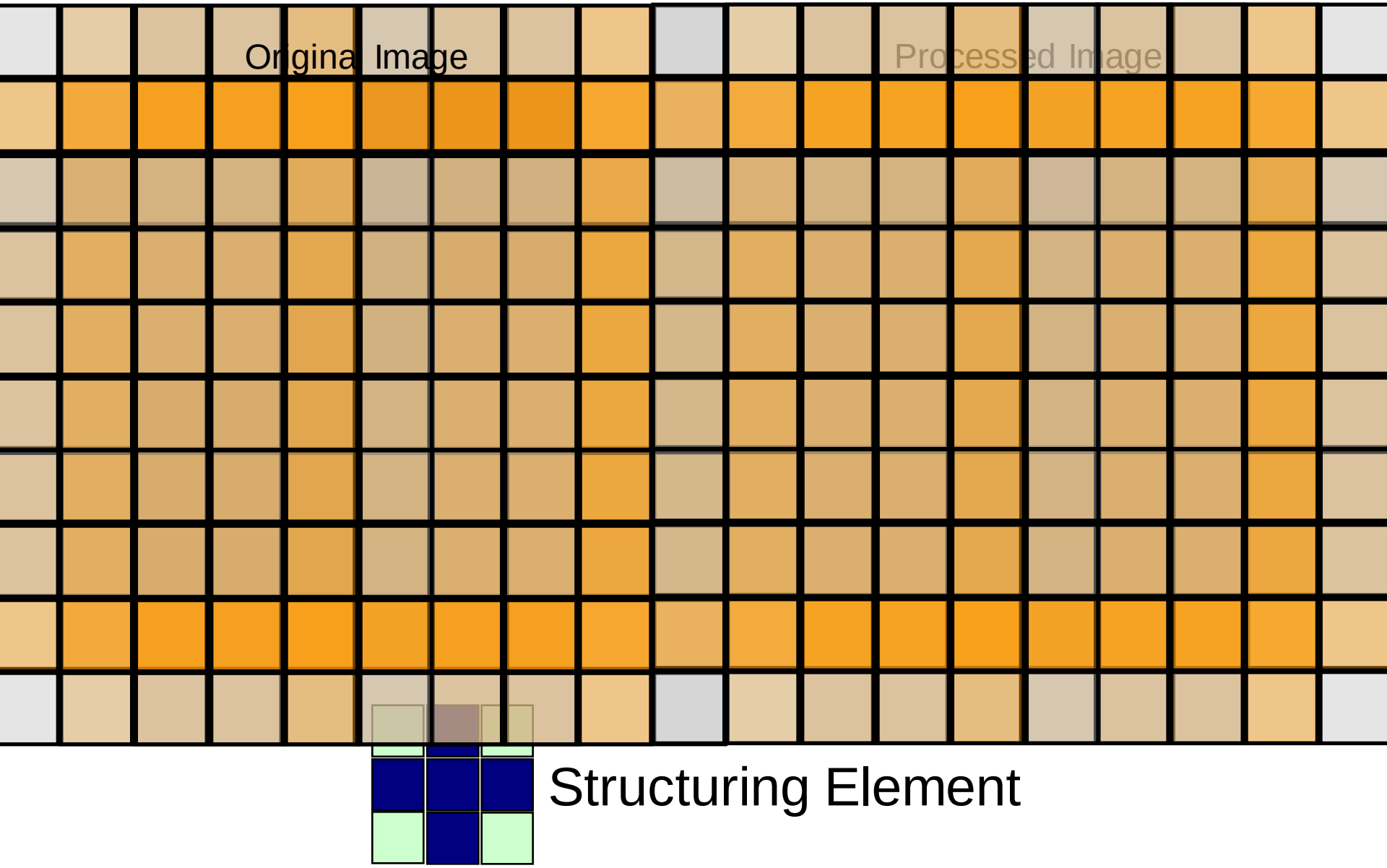


Image After Opening



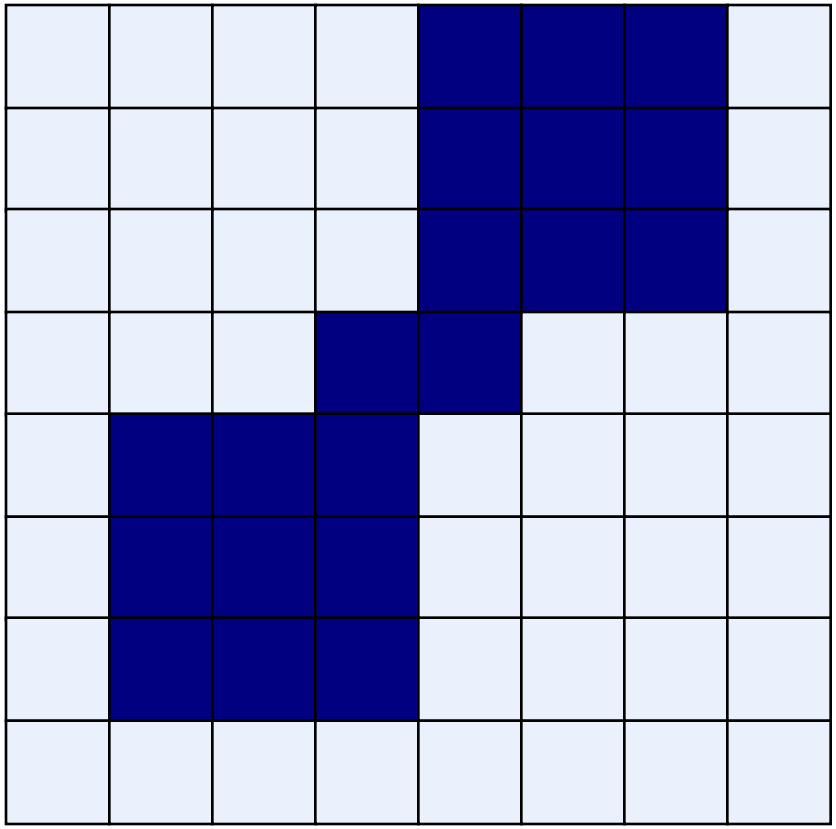
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Opening Example

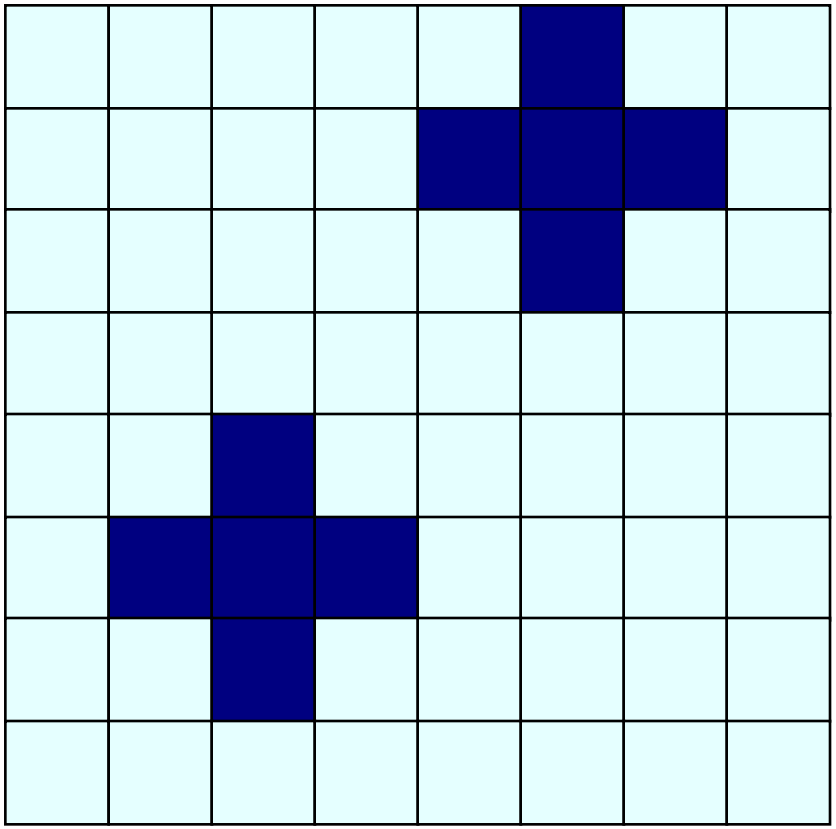


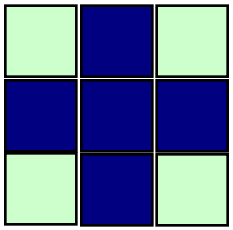
Opening Example

Original Image



Processed Image



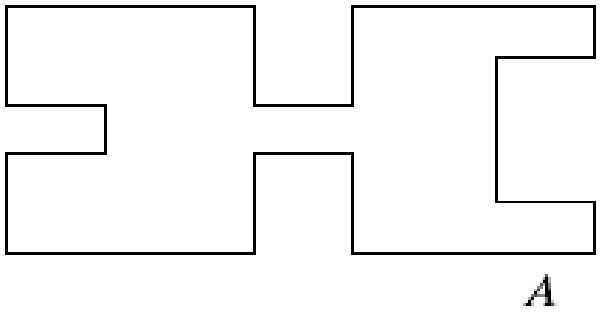


Structuring Element

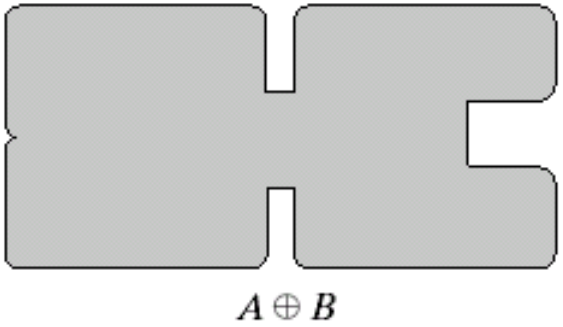
Closing

The closing of image f by structuring element s, denoted $f \bullet s$ is simply a dilation followed by an erosion.

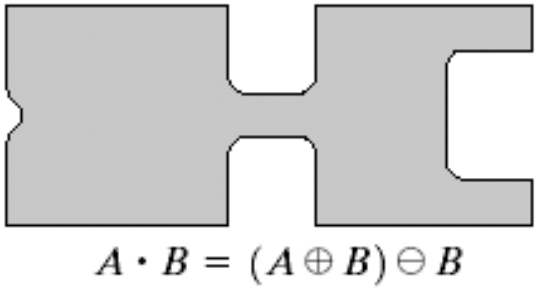
$$f \bullet s = (f \oplus s) \ominus s$$



Original shape



After dilation



After erosion (closing)

A disc shaped structuring element is used.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Closing Example

Original Image

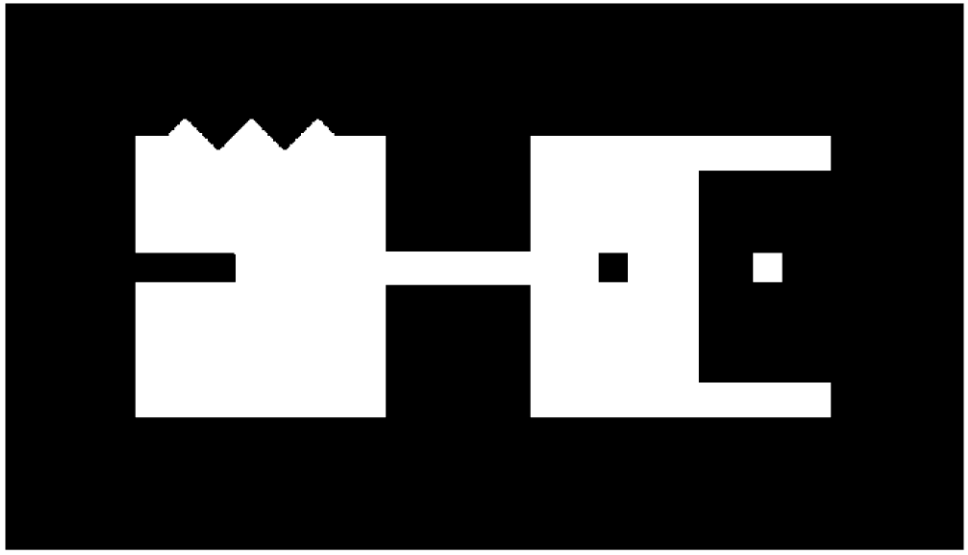
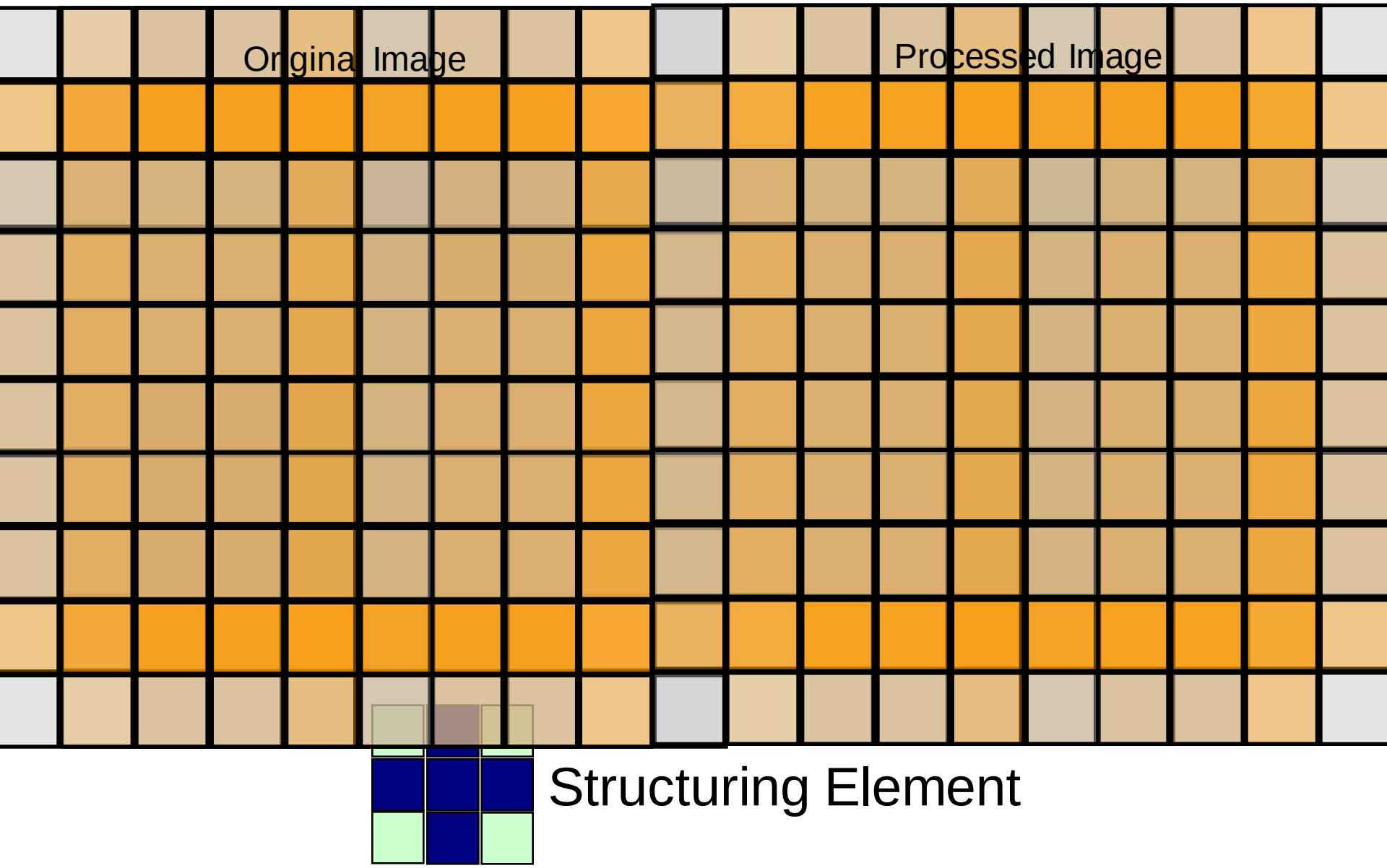


Image After Closing



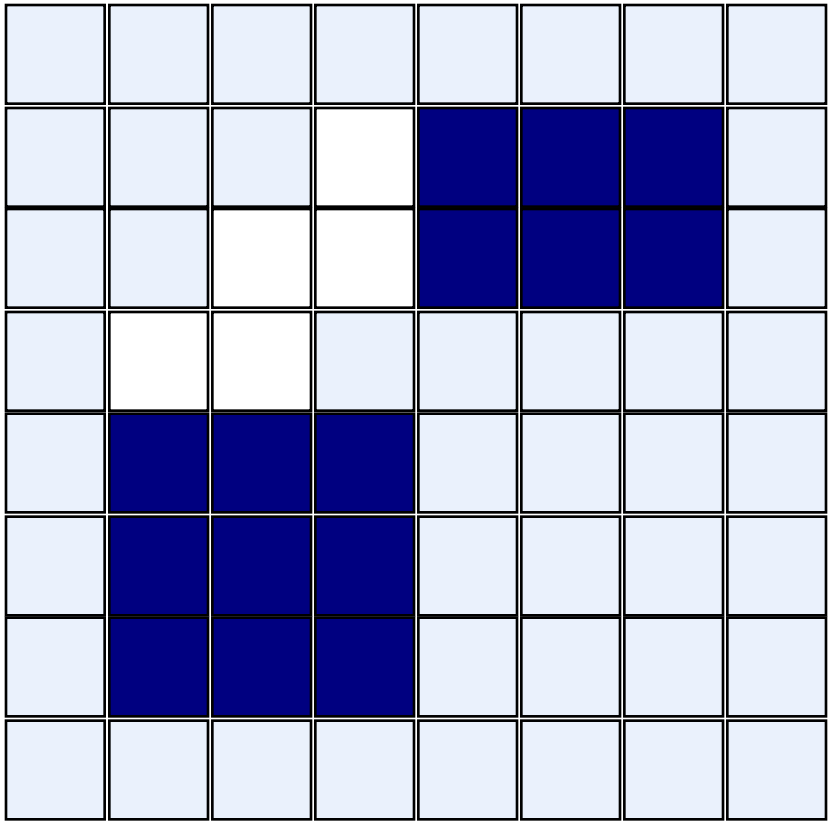
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Closing Example

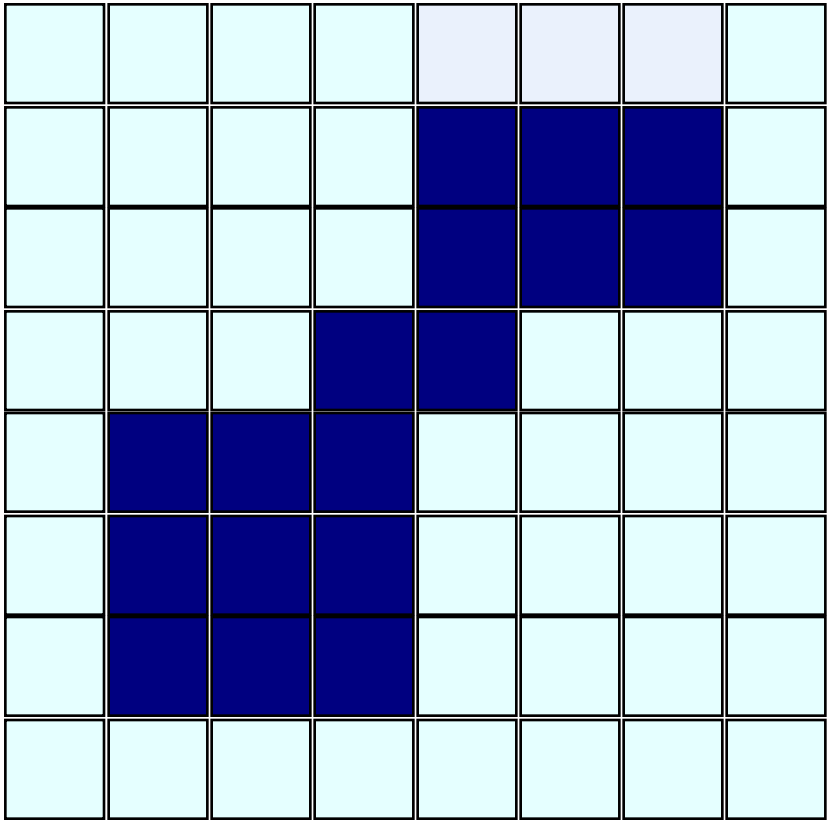


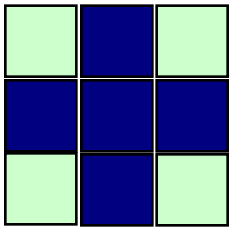
Closing Example

Original Image



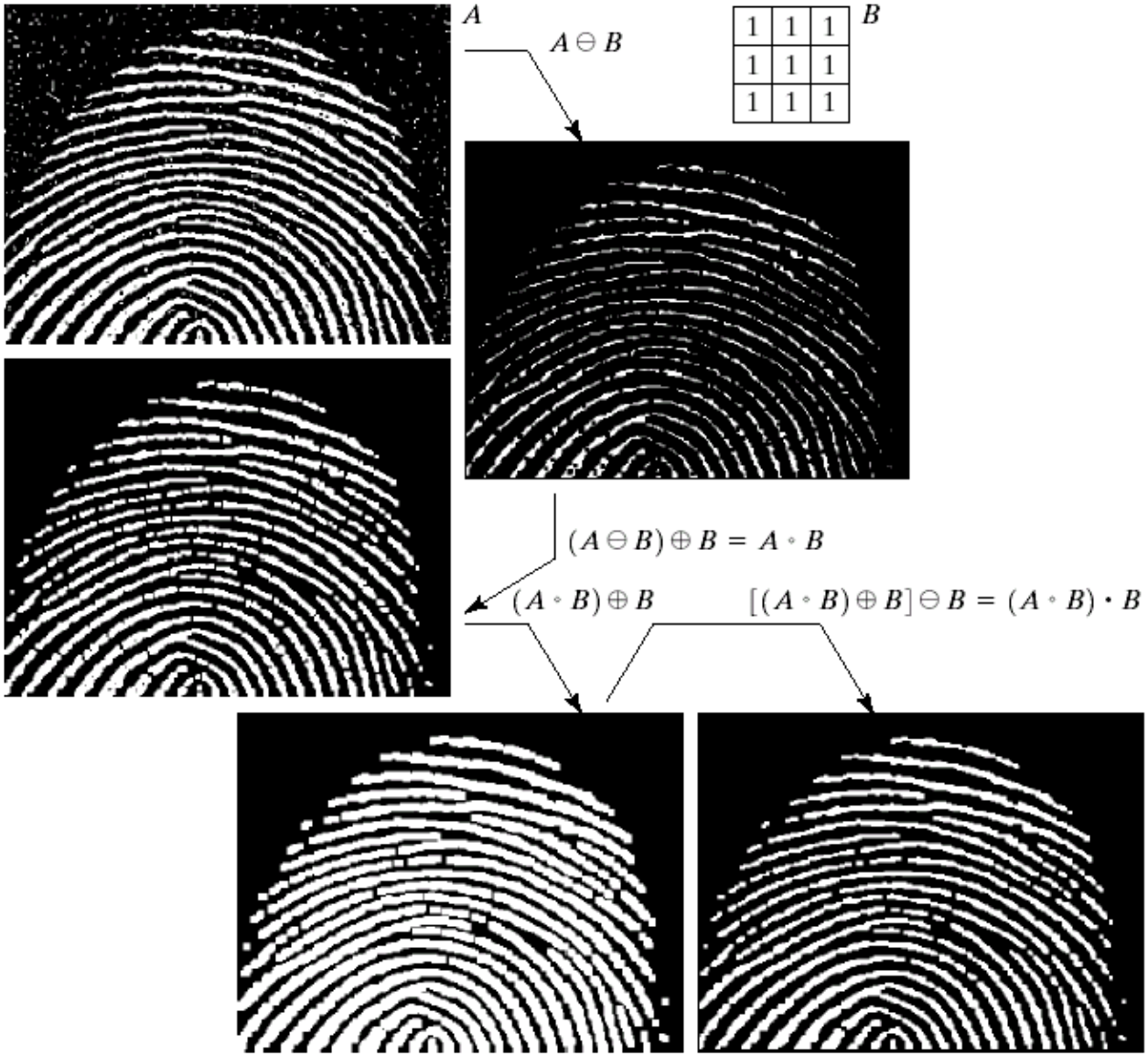
Processed Image





Structuring Element

Morphological Processing Example



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Morphological Algorithms

Using these simple technique we can perform some more interesting morphological algorithms.

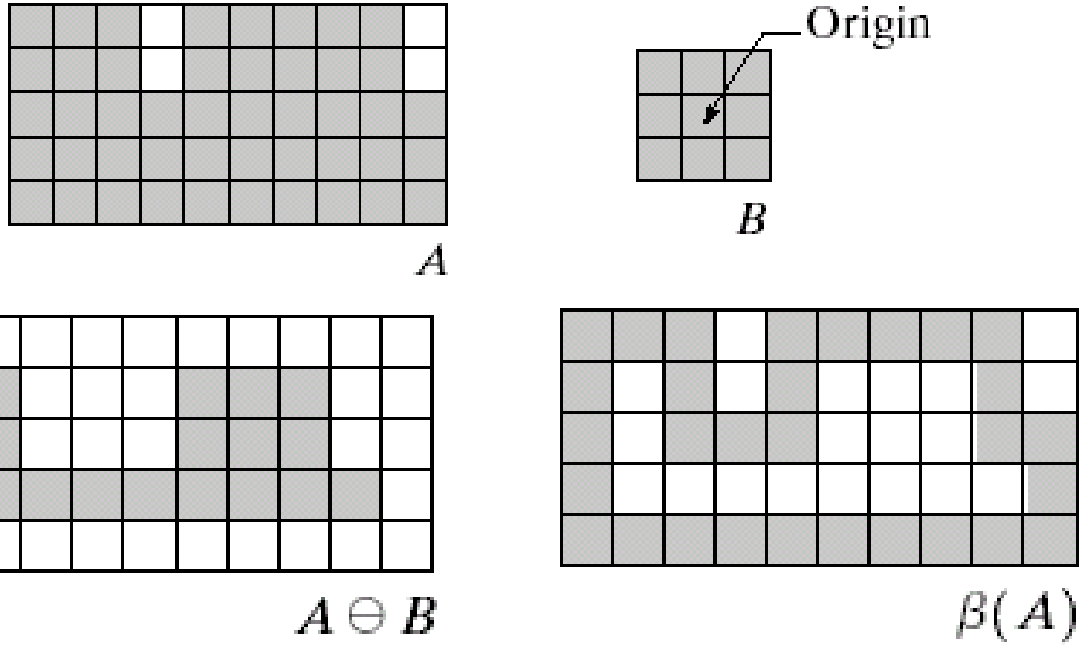
- Boundary extraction
- Region filling
- Extraction of connected components
- Thinning/Skeletonization

Boundary Extraction Algorithm

Extracting the boundary (or outline) of an object is often extremely useful.

The boundary can be given simply as

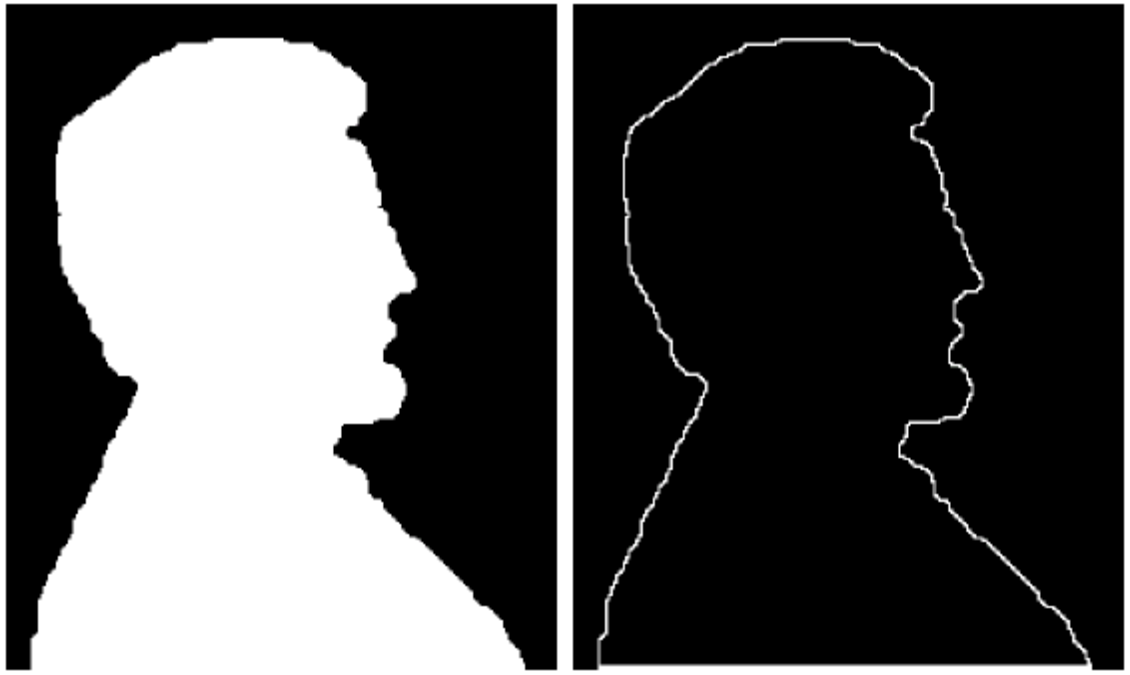
$$\beta(A) = A - (A \ominus B)$$



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Boundary Extraction Example

A simple image and the result of performing boundary extraction using a square 3x3 structuring element.

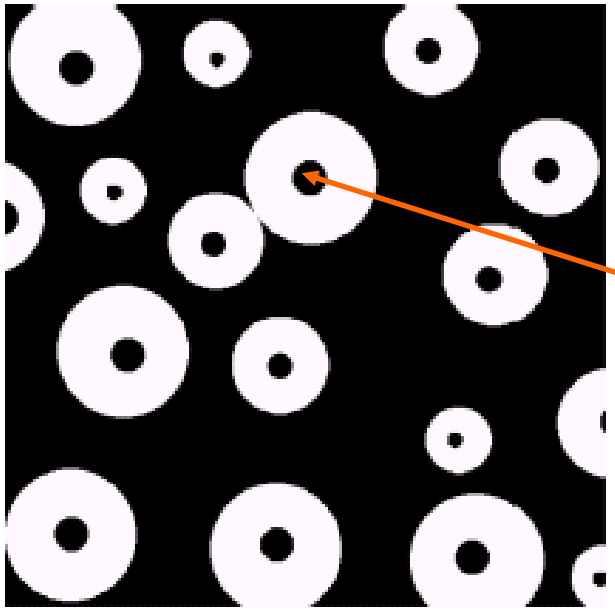


Original Image Extracted Boundary

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Region Filling

Given a pixel inside a boundary, region filling attempts to fill that boundary with object pixels.



Given a point inside here, can we fill the whole circle?

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Region Filling Algorithm

1. The key equation for region filling is

$$X_k = (X_{k-1} \oplus B) \cap A^c \qquad k = 1,2,3.....$$

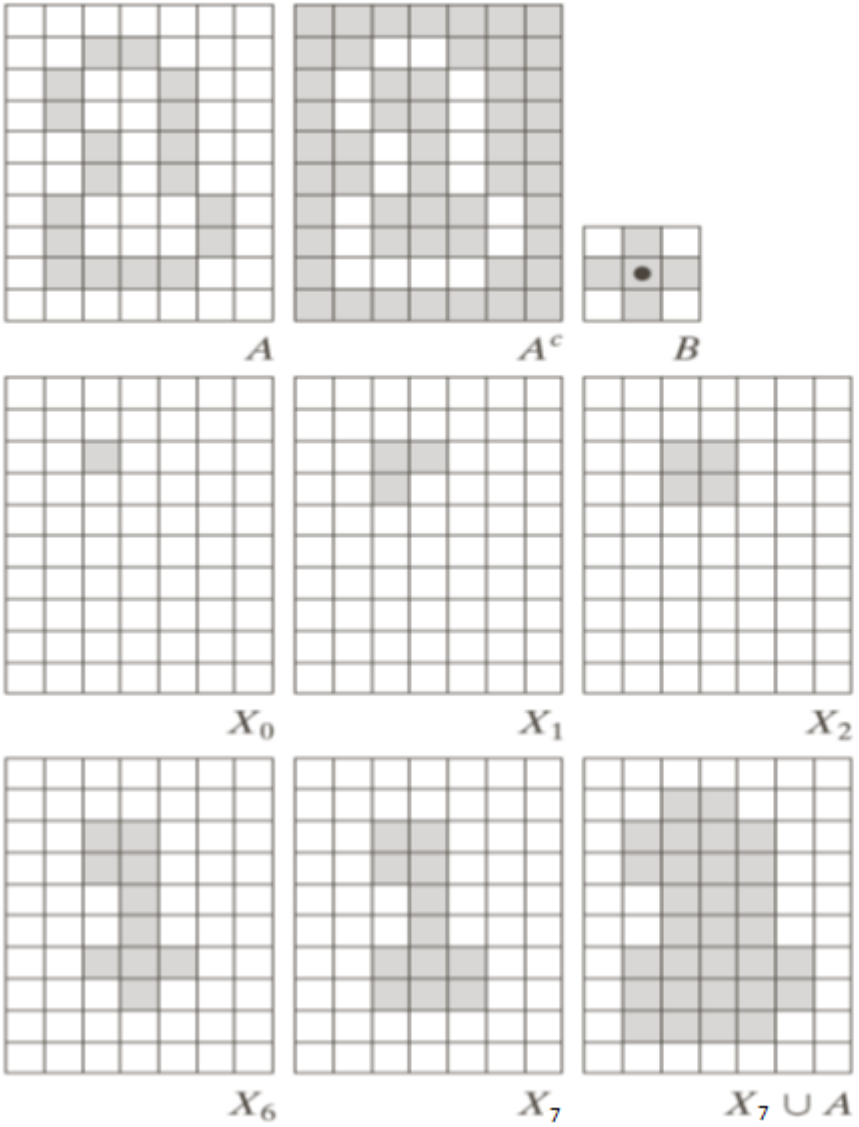
Where X_0 is simply the starting point inside the boundary, B is a simple structuring element and A^c is the complement of A.

2. This equation is applied repeatedly until X_k is equal to X_{k-1}.

3. Finally the result is unioned with the original boundary.

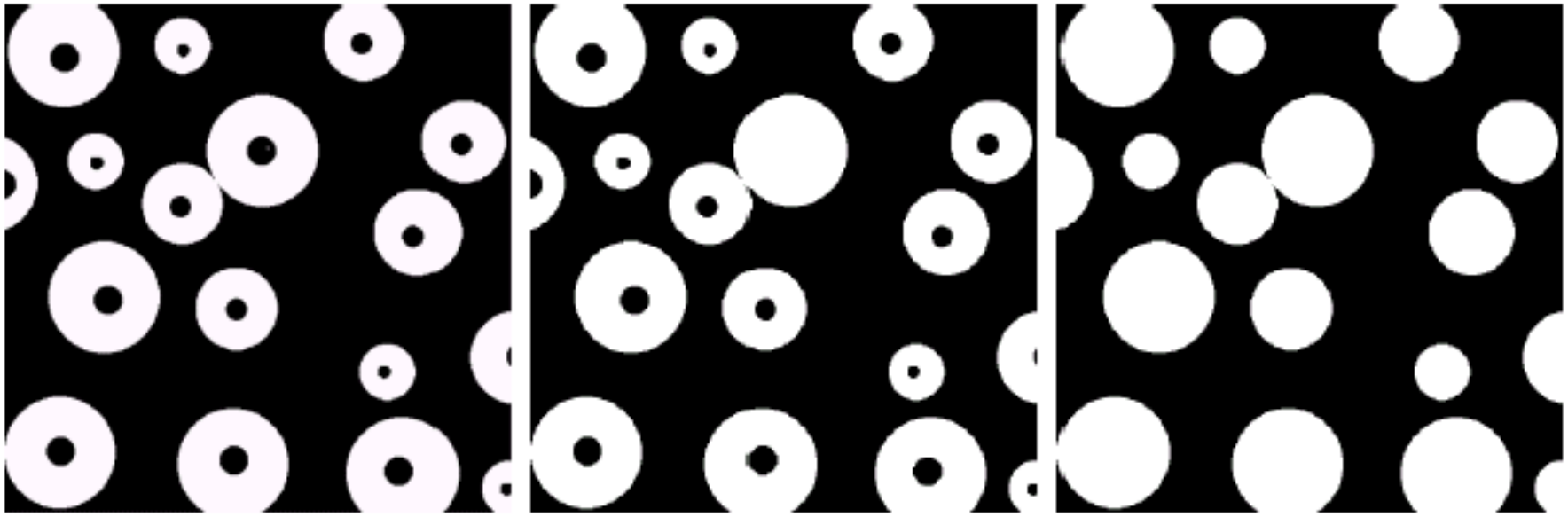
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Region Filling Step By Step



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Region Filling Example



Original Image

One Region Filled

All Regions Filled

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Extraction of Connected Component Algorithm

1. The key equation for connected component extraction is

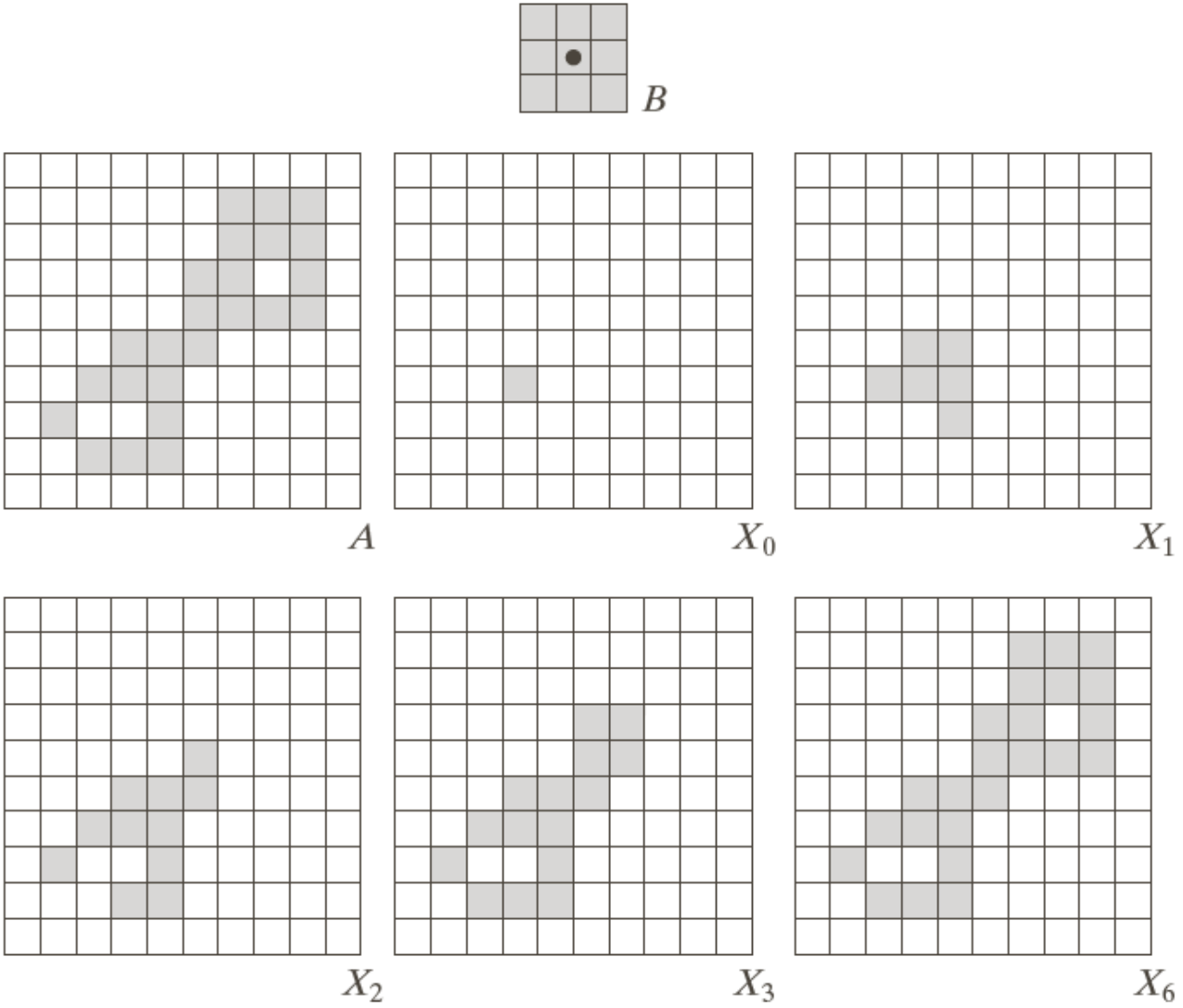
$$X_k = (X_{k-1} \oplus B) \cap A \qquad k = 1, 2, 3.....$$

Where X_0 is simply the starting point of the connected component which is known to us beforehand and B is a simple structuring element.

2. This equation is applied repeatedly until X_k is equal to X_{k-1}.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Extraction of Connected Component Step By Step

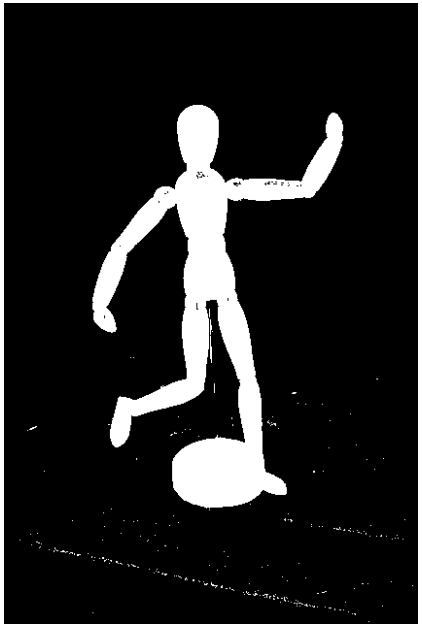


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

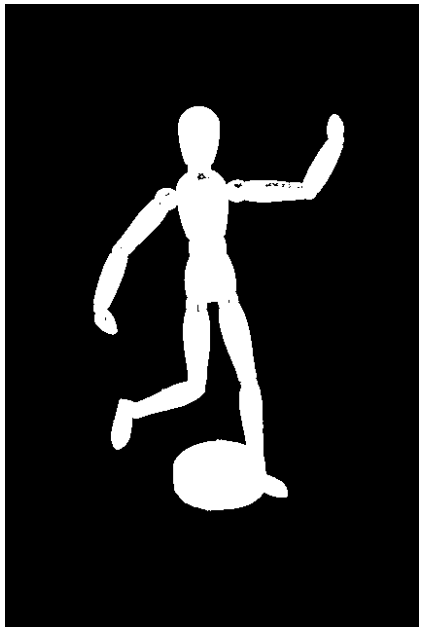
Improving Segmentation Outcome



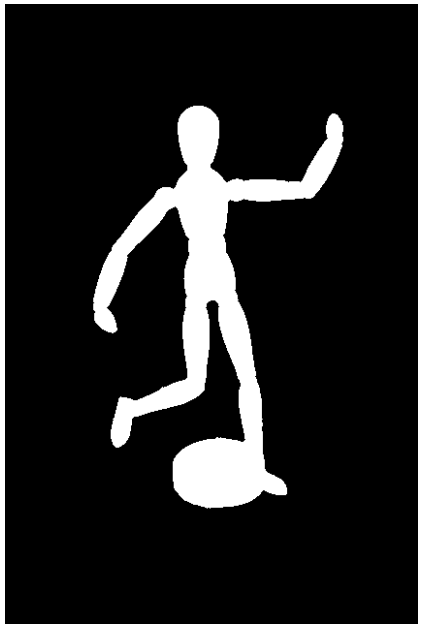
Original Image



Initial thresholding



After Opening

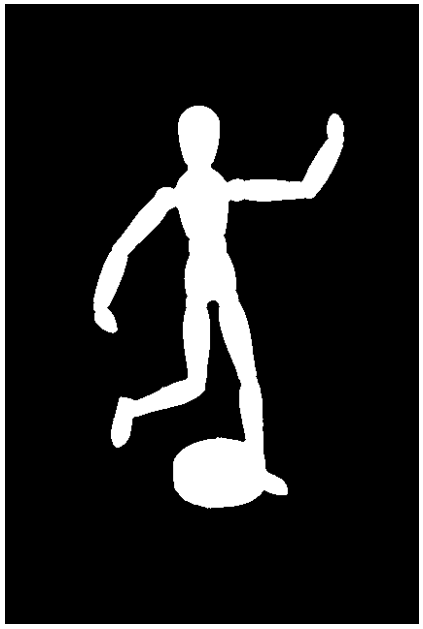


After Closing

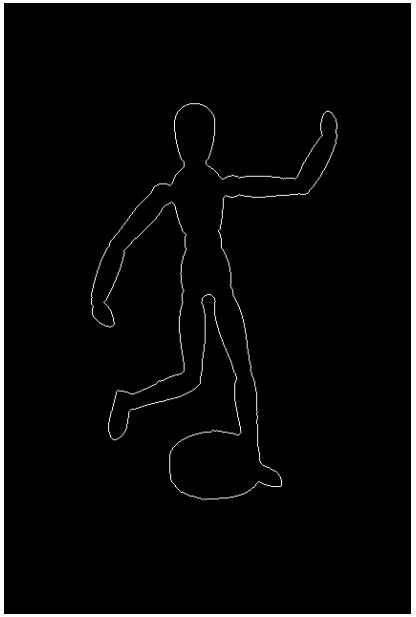
Thinning / Skeletonization Example



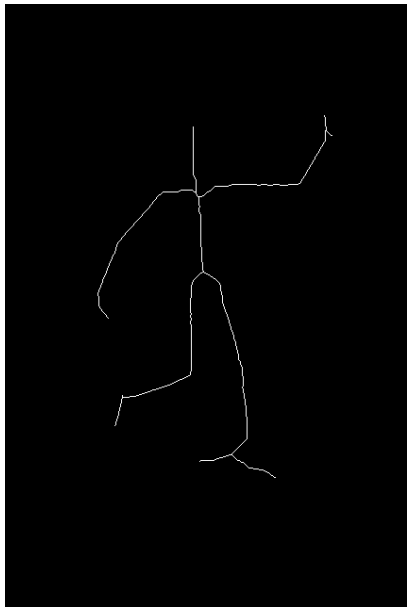
Original Image



Segmented Image



Boundary Extraction



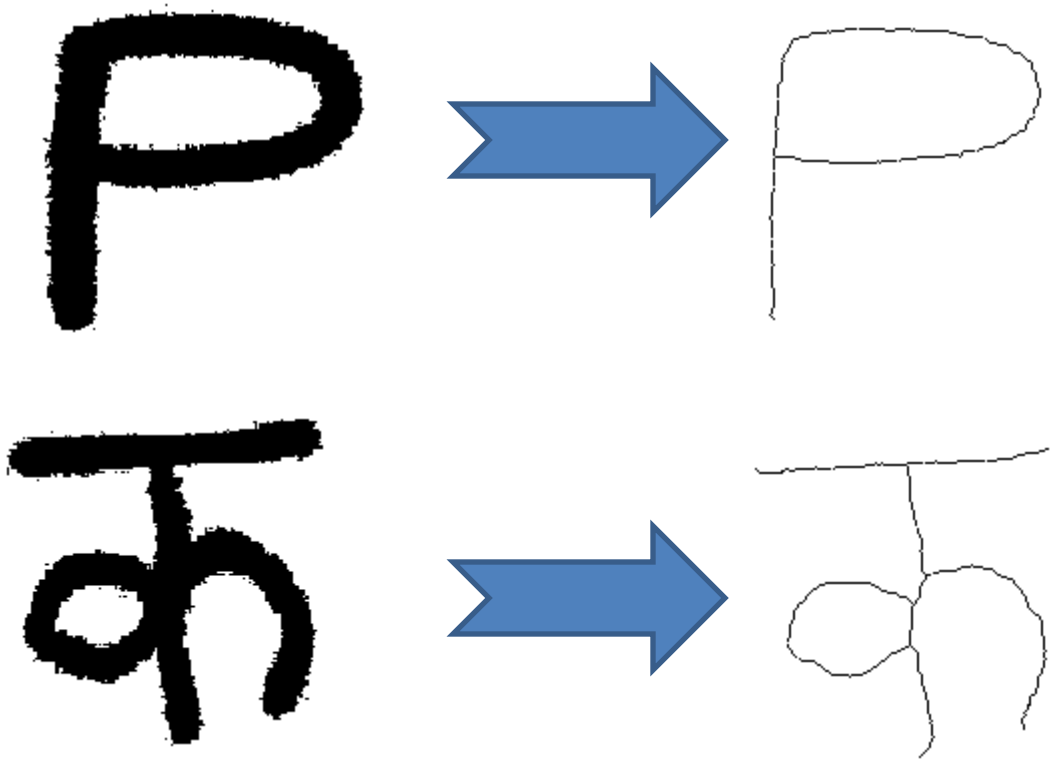
Thinned Image

Thinning

Generates single pixel wide skeletons of the input images.

It has wide applications in shape analysis and Document image processing.

Thinning / Skeletonization Example



Original Image

Repeatedly run until 1-pixel thick

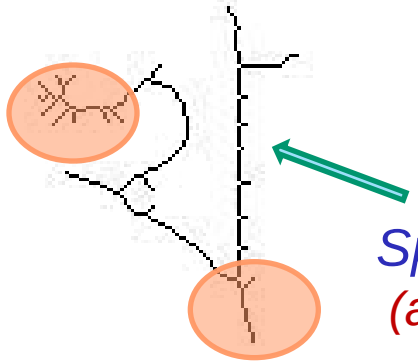
Why Thinning?

- Digital skeleton of character images has a wide range of applications for shape analysis and classification.

- Most of the methods can not preserve the local features and true shapes of images at the junction point and end point.

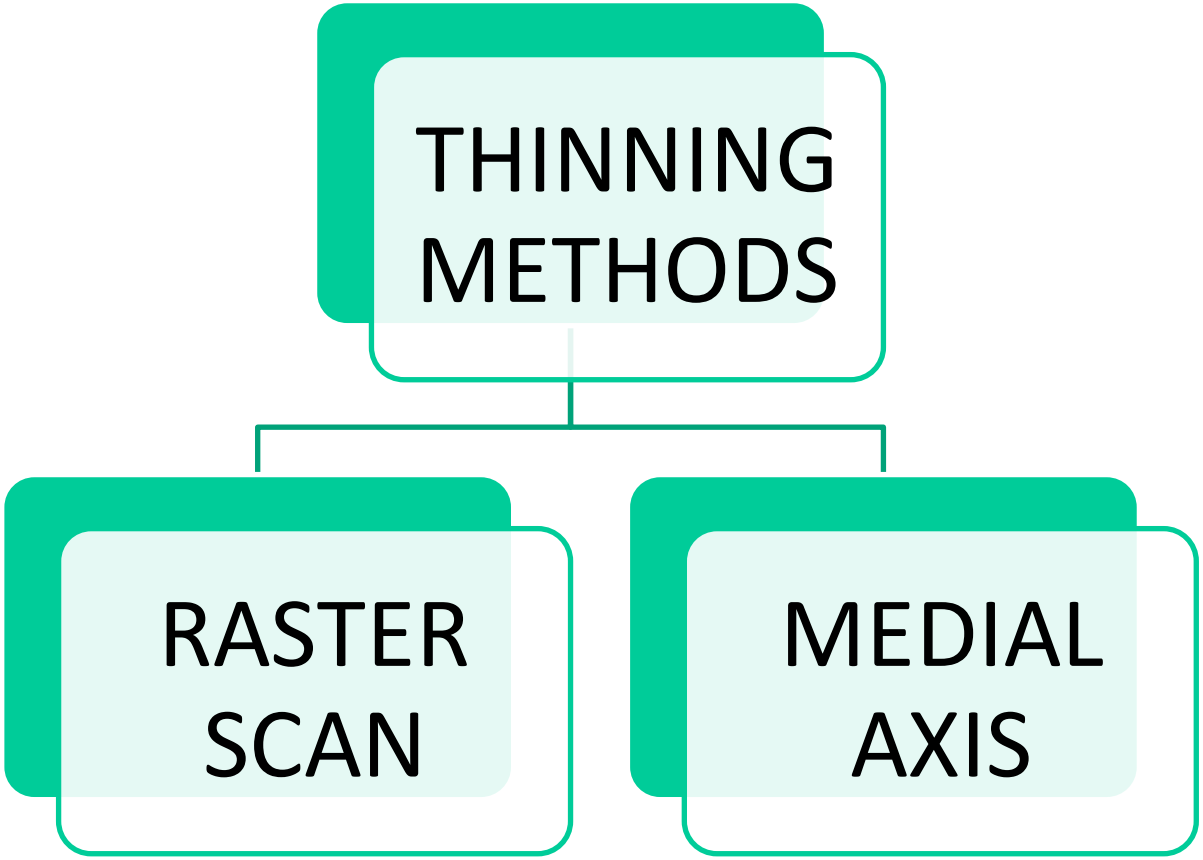
Shape distortion at Junction and End Points (circle)





Spurious strokes (arrow)

Thinning Methods



Raster Scan Based Methods

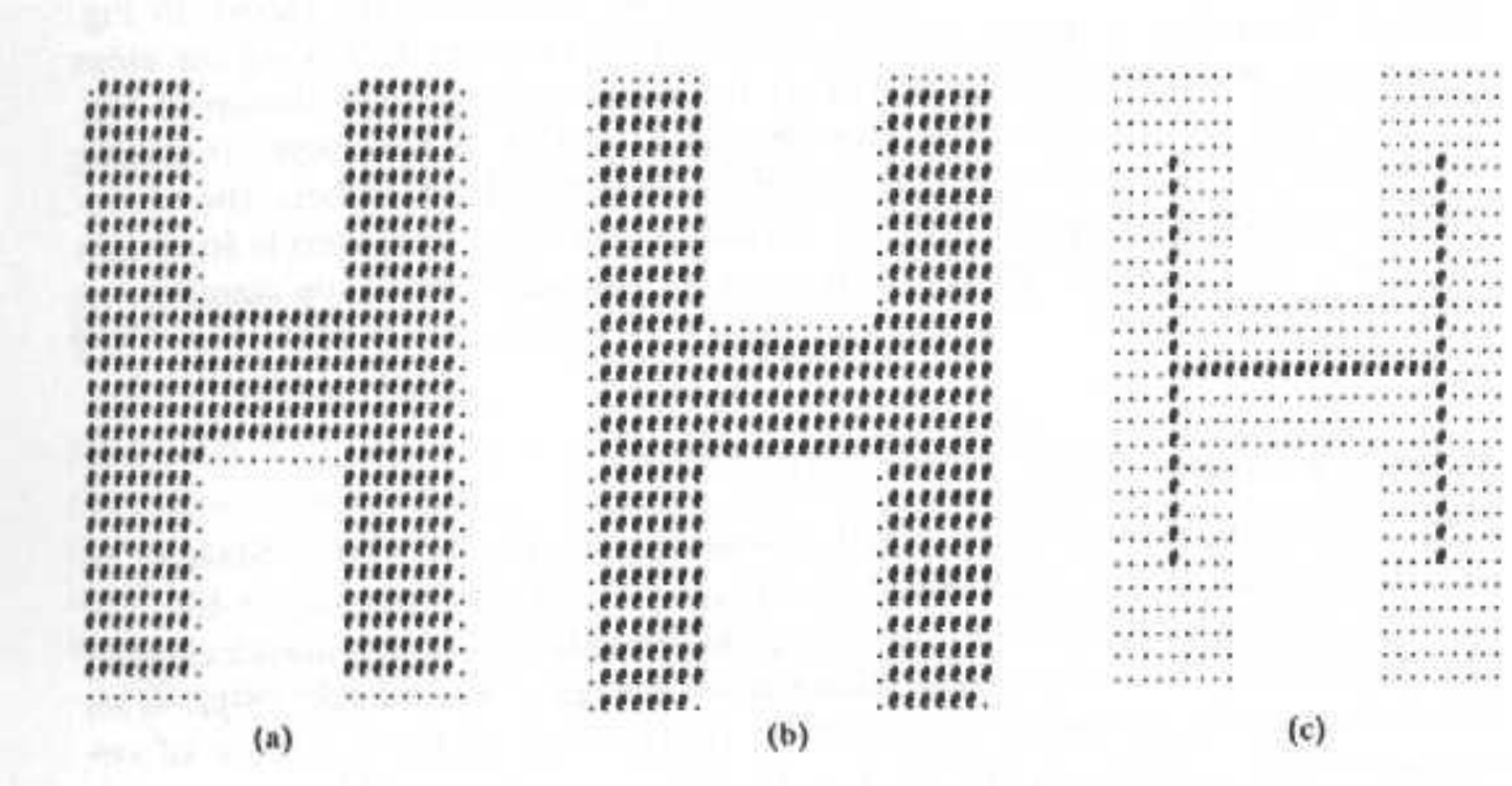
They are rule based methods.

Uses templates to eliminate pixels.

If the neighborhood of the pixel matches the mask/template then the pixel is removed.

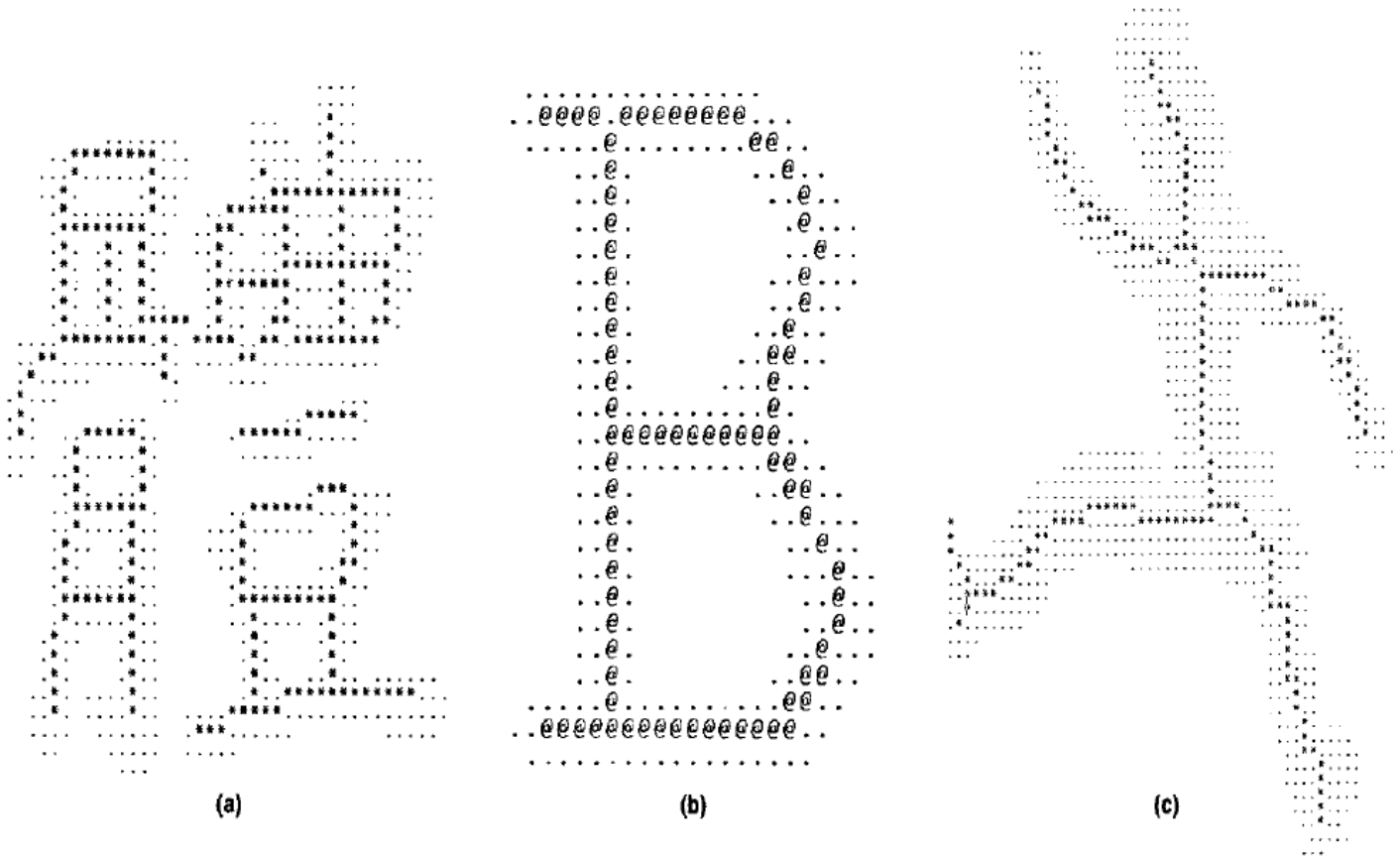
This mask is run over the entire image and the process is repeated for each pixel in the image.

Raster Scan Based Thinning / Skeletonization Example



T.Y. Zhang, C.Y. Suen, *A fast parallel algorithm for thinning digital patterns*, Communications of the ACM, vol. 27, No. 3, pp. 236-239, 1984.

Raster Scan Based Thinning / Skeletonization Example



T.Y. Zhang, C.Y. Suen, *A fast parallel algorithm for thinning digital patterns*, Communications of the ACM, vol. 27, No. 3, pp. 236-239, 1984.

Raster Scan Based Thinning / Skeletonization Example

1	1	0
0	1	0
0	1	1

0	1	0
1	1	0
0	0	1

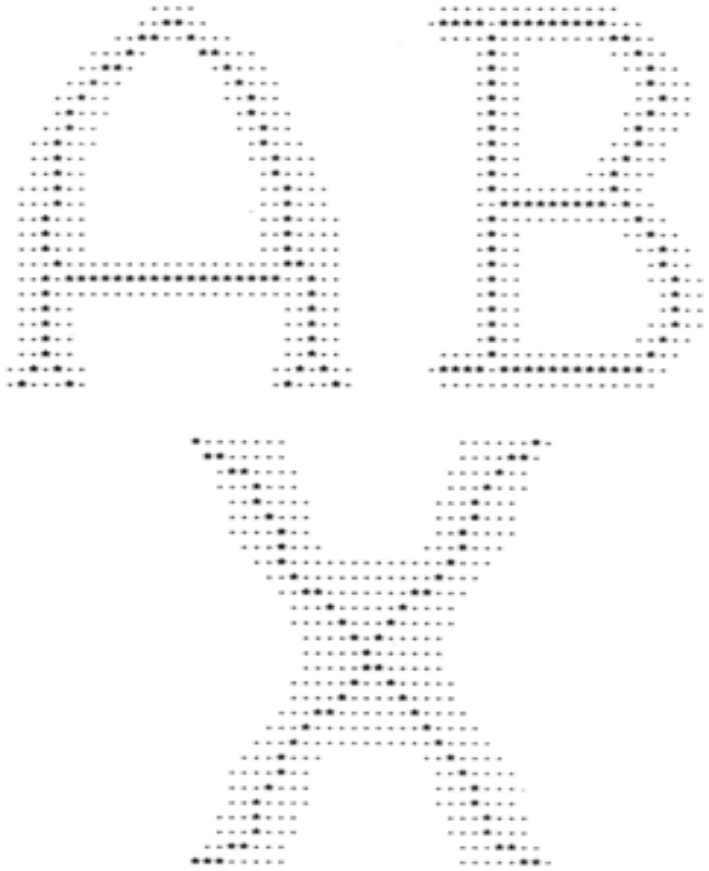
1	1	1
0	1	0
1	1	1

0	1	0
1	1	1
0	1	0

Templates for pixel preservation

A. Datta and S. K. Parui, *A robust parallel thinning algorithm for binary images*, Pattern Recognition, Elsevier, vol. 27, pp. 1181-1192, 1994.

Raster Scan Based Thinning / Skeletonization Example



A. Datta and S. K. Parui, *A robust parallel thinning algorithm for binary images*, Pattern Recognition, Elsevier, vol. 27, pp. 1181-1192, 1994.

Medial Axis based Methods

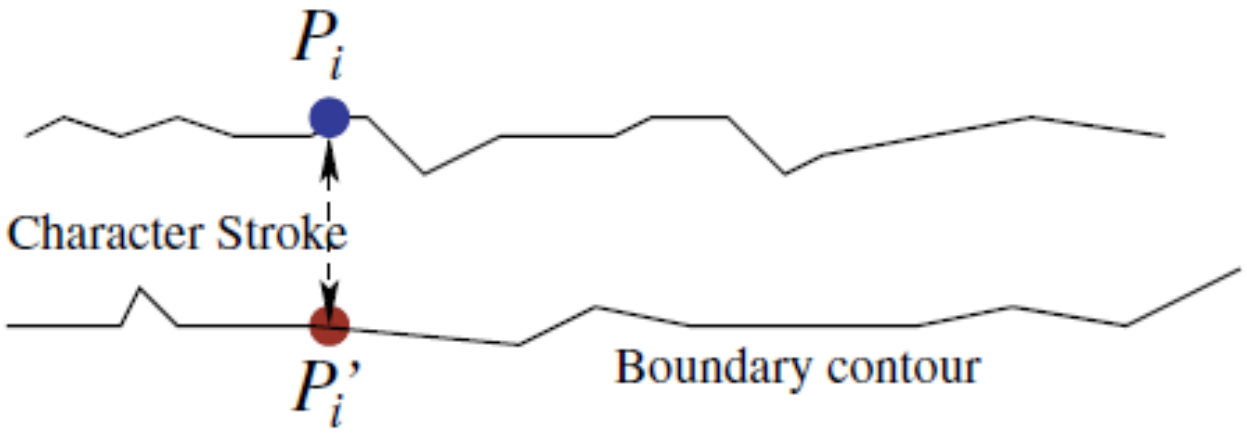
They generate medial points in between two parallel contour segments.

This process is repeated for all parallel contour segments.

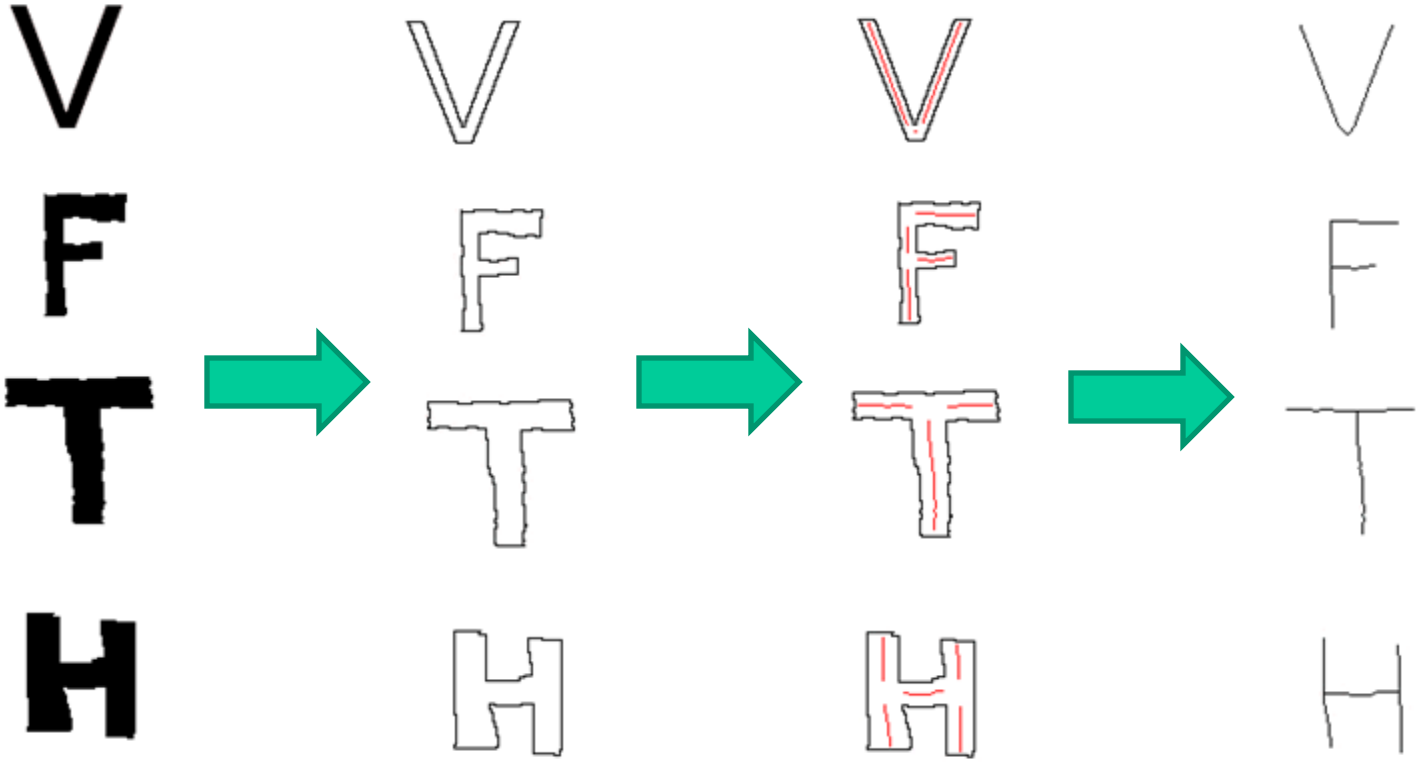
Interpolation may be used to reconstruct the junction regions.

Medial-axis Based Methods

Medial Axis Generation



Medial-axis Based Thinning / Skeletonization Example



S. Bag and G. Harit, *Skeletonizing character images using a modified medial axis-based strategy*, IJPRAI, vol. 25, No. 7, pp. 1035-1054, 2011.

Mathematical Example

Problem : Consider a binary image **S,** which is the result of graylevel segmentation on image **I** as shown below. The object pixels in image **S** are labelled as ‘1’.

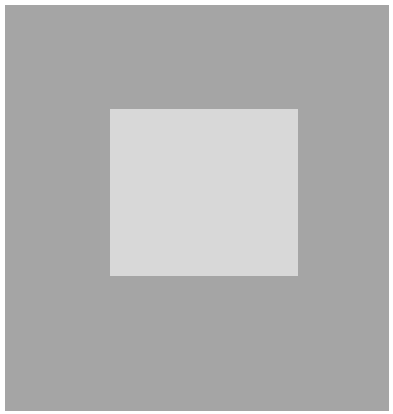


Image I

Gray-level Segmentation

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0
0	1	0	1	1	1	0	1	0
0	0	0	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0

Image S

Mathematical Example

If you apply Labeling Algorithm on image **S**, how many objects will be detected ?

Answer: *Labeling algorithm - 5 objects*

How many object do you observe visually in image **I** ?

Answer: *Observe visually – 1 object*

How to solve this conflict?

Use *Opening* (*Erosion* followed by *Dilation*) to remove small objects treated as noise.

Summary

The purpose of morphological processing is primarily to remove imperfections added during segmentation.

The basic operations are *erosion* and *dilation.*

Using the basic operations we can perform *opening* and *closing.*

More advanced morphological operation can then be implemented using combinations of all of these.